

AI Deep Learning Framework for Vietnamese Multi-Speaker Transcription and Identification in Noisy Environments

GROUP AND MEMBER INFORMATION

GROUP ID: GSP26AI21

PROJECT CODE: SP26AI91

MEMBER:

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Role

Member	Main Task	Specific Responsibilities
Sang	Tool Development and Manual Labeling	Find and build the labeling tool; write supporting scripts for labeling; manually label the data.
Duy	Documentation and Manual Labeling	Write the report and technical documentation; manually label the data together with Member 1.
Phuong	Synthetic Data and ASR	Create the synthetic dataset; work on the ASR component and the Qwen model.
Khang	Pipeline and Diarization	Build the main processing pipeline; work on the two diarization models: DiariZen and Pyannote.

Presentation Outline



Introduction

Context & Problem



Data

Corpus & Augment



Architecture

Model Pipeline



Metrics

Evaluation KPIs



Training

Parameters & Env



Results

Performance Data



Demo

System Showcase



Conclusion

Scalability

1 - INTRODUCTION

1.1 Background & Motivation

- Multi-speaker meeting transcription is difficult.
- Key challenges: overlap, interruptions, noise, reverberation, and recording variability.
- For Vietnamese, the task is harder due to tonal language properties, regional pronunciation variation, and informal or mixed-language speech.



Tonal Language



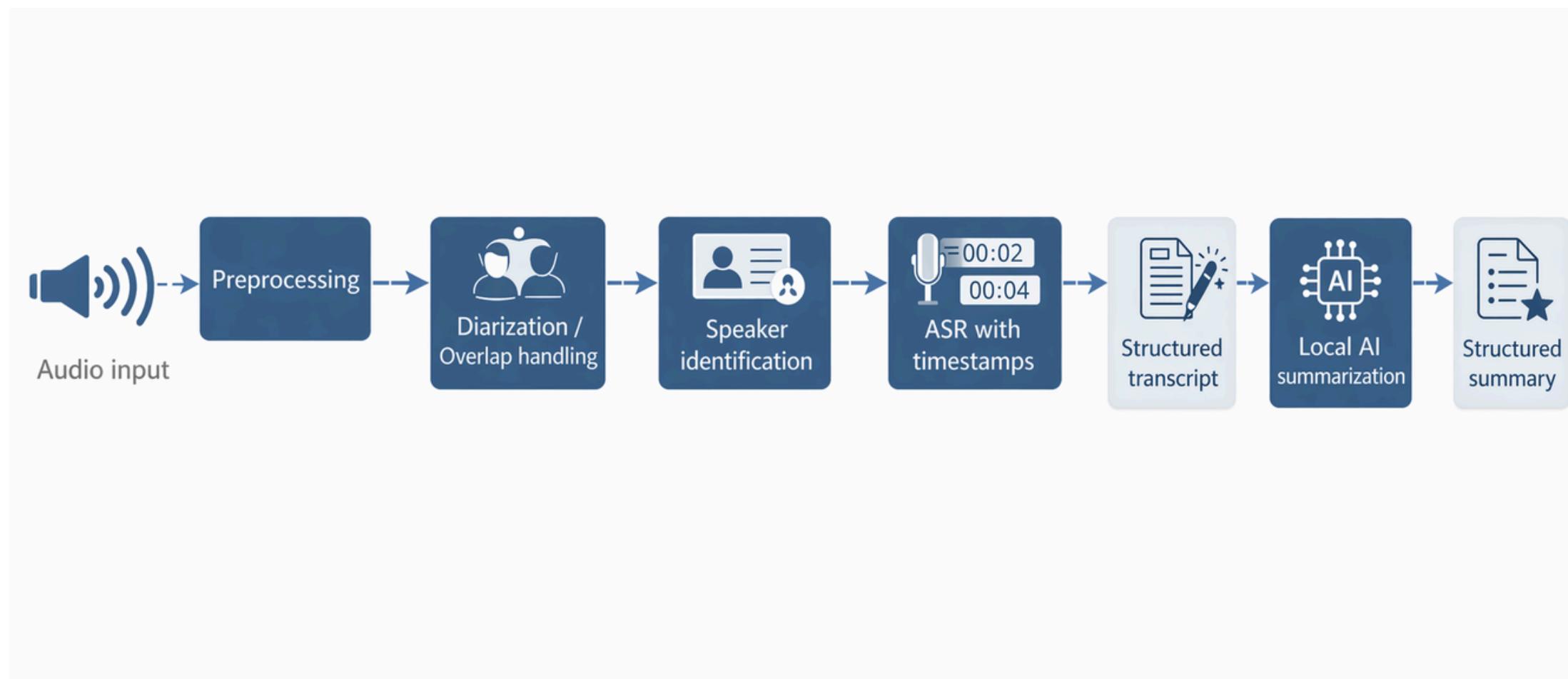
Background Noise



Speaker Overlap

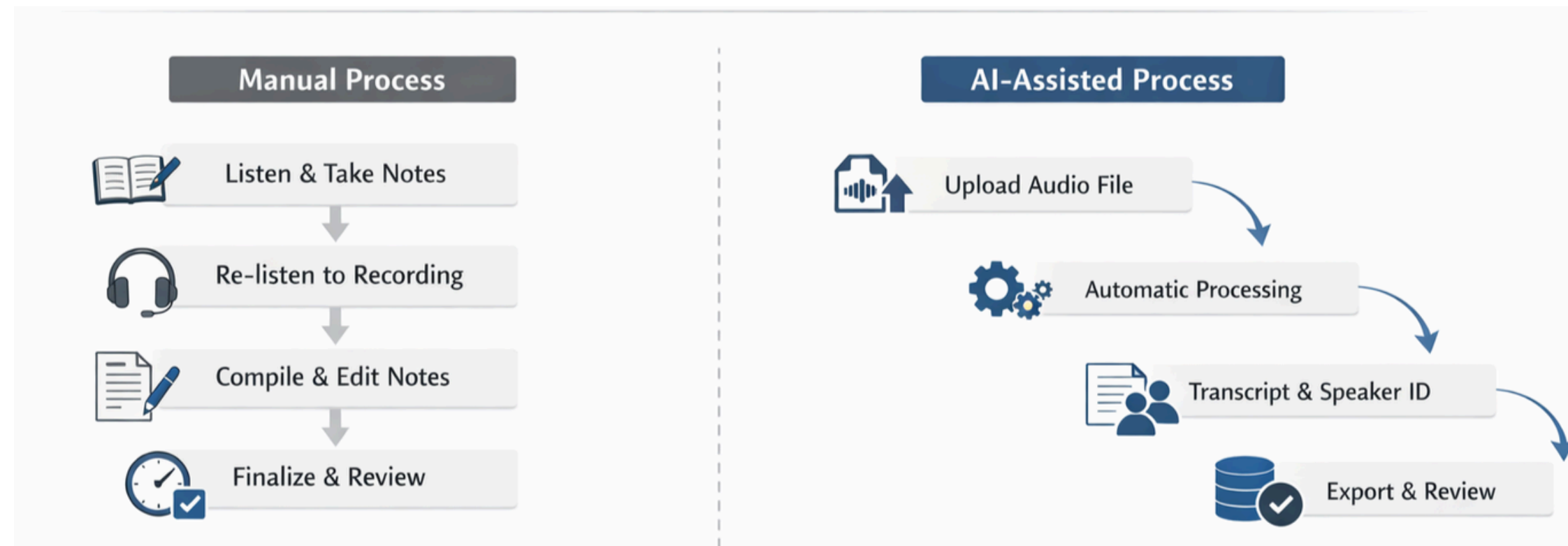
1.2 Objective

- Perform **speaker diarization**, **speaker identification** and **speech recognition** under noisy and overlap-prone conditions.
- Produce a timestamped transcript that preserves speaker identity information as accurately as possible.
- Using local AI to Summary transcript and decrease time to understand conversation.



1.3 Project Aim

- The project aims to deliver both a research contribution and a practical prototype.
- At the research level, it investigates how multiple deep learning components can be integrated to address Vietnamese overlapping and noisy multi-speaker speech.
- At the application level, it targets a usable system that reduces manual note-taking effort and converts spoken meeting content into structured, searchable, and actionable information.



1.4 Applicability

Positioned as a meeting intelligence tool that:

- Automates note-taking and minutes generation
- Converts meeting content into searchable digital assets
- Supports downstream generation of meeting minutes and action items

2 - Data

2.1 Dataset Overview

Table 8: Dataset usage overview in the proposed system.

Task	Purpose	Source	Usage
ASR	Speech recognition training	VIVOS, Common Voice	Train / adapt ASR
Diarization	Speaker segmentation and overlap handling	Synthetic meetings, self-labeled data	Train / adapt Diarization
Evaluation	Real-world testing	Collected meeting recordings	Test robustness in real audio
Noise robustness	Noisy audio conditions	Augmented / noisy meetings	Evaluate denoising impact

2.1 Dataset Overview

Public Datasets

Table 10: Public Vietnamese speech corpora used as source data for synthetic data generation.

Dataset	Speakers	Synthetic Data Contribution	Sampling Rate	Usage
VIVOS	46	Part of the 210 h synthetic training set; 33.19 h validation set	16,000 Hz	Training and validation for diarization and ASR
LSVSC	6	Part of the 210 h synthetic training set	16,000 Hz	Training for diarization and ASR
Common Voice	6	Part of the 210 h synthetic training set	16,000 Hz	Training for diarization and ASR
Speech Noise dataset	0	947 noise segments	16,000 Hz	Acoustic augmentation and robustness testing

2.1 Dataset synthetic

Table 25: General statistics of the training and validation datasets.

Attribute	Train Set	Val Set
Total speakers	18 (6 VIVOS + 6 LSVSC + 6 CV)	40 (VIVOSSPK)
Total audio duration	210.34 hours	33.19 hours
Number of meetings	399 meetings	63 meetings
Duration per meeting	30-35 minutes	28-35 minutes
Ratio of meetings with noise	202/399 (50.6%)	32/63 (50.8%)
Ratio of meetings with reverb	167/399 (41.9%)	20/63 (31.7%)
Noise SNR range	20-35 dB	20-35 dB
Reverb RT60 range	0.15-0.4 s	0.15-0.4 s
Participants per meeting	2, 3, or 4	2, 3, or 4
Overlap levels	0%, 5%, 10%, 15%, 20%, 25%, 30%	0%, 5%, 10%, 15%, 20%, 25%, 30%
Sample rate	16000 Hz, mono	16000 Hz, mono
Label formats	WAV, RTTM, TXT, STM, CTM, JSON	WAV, RTTM, TXT, STM, CTM, JSON
Noise sources	947 WAV files	947 WAV files

Table 18: Room simulation parameters.

Parameter	Range	Unit
Room width	4.0 - 10.0	m
Room length	5.0 - 12.0	m
Ceiling height	2.5 - 4.0	m
RT60 (reverb time)	0.15 - 0.4	s
Speaker-microphone radius	1.5 - 2.5	m
Probability of applying RIR	40%	

2.1 Dataset synthetic

Train:

Overlap Level	Meetings	Hours	Noise	Reverb	2spk	3spk	4spk
overlap_0	57	29.90 h	34/57 (60%)	25/57 (44%)	19	19	19
overlap_5	57	30.09 h	26/57 (46%)	24/57 (42%)	19	19	19
overlap_10	57	30.02 h	22/57 (39%)	21/57 (37%)	19	19	19
overlap_15	57	30.20 h	28/57 (49%)	23/57 (40%)	19	19	19
overlap_20	57	29.84 h	29/57 (51%)	22/57 (39%)	19	19	19
overlap_25	57	29.96 h	29/57 (51%)	24/57 (42%)	19	19	19
overlap_30	57	30.34 h	34/57 (60%)	28/57 (49%)	19	19	19
TOTAL	399	210.34 H	202 (50.6%)	167 (41.9%)	133	133	133

Val:

Overlap Level	Meetings	Hours	Noise	Reverb	2spk	3spk	4spk
overlap_0	9	4.68 h	4/9 (44%)	2/9 (22%)	3	3	3
overlap_5	9	4.78 h	5/9 (56%)	4/9 (44%)	3	3	3
overlap_10	9	4.73 h	5/9 (56%)	3/9 (33%)	3	3	3
overlap_15	9	4.79 h	6/9 (67%)	3/9 (33%)	3	3	3
overlap_20	9	4.77 h	3/9 (33%)	5/9 (56%)	3	3	3
overlap_25	9	4.55 h	6/9 (67%)	3/9 (33%)	3	3	3
overlap_30	9	4.89 h	3/9 (33%)	0/9 (0%)	3	3	3
TOTAL	63	33.19 H	32 (50.8%)	20 (31.7%)	21	21	21

2.1 Dataset synthetic

Synthetic dataset:

1. Select speaker clips

- Short utterances are sampled from source datasets.

2. Compose multi-speaker meetings

- Clips are concatenated into meetings with 2–4 speakers.

3. Add overlap and acoustic effects

- Overlap, noise, and reverberation are injected to simulate real meetings.

4. Export aligned ground truth

- RTTM, STM/CTM, speaker labels, and metadata are generated automatically.

Meeting length: 30–35 min

Speakers per meeting: 2–4

Overlap conditions: 5%,10%,15%

Output labels: RTTM / STM / Transcript / Speaker metadata

noise_only	
1.wav	2437 noise_only\10.wav,noise_only
2.wav	2438 noise_only\100.wav,noise_only
3.wav	2439 noise_only\101.wav,noise_only
4.wav	2440 noise_only\102.wav,noise_only
5.wav	2441 noise_only\103.wav,noise_only
6.wav	2442 noise_only\104.wav,noise_only
7.wav	2443 noise_only\105.wav,noise_only
8.wav	2444 noise_only\106.wav,noise_only
9.wav	2445 noise_only\107.wav,noise_only
10.wav	2446 noise_only\108.wav,noise_only
11.wav	2447 noise_only\109.wav,noise_only
12.wav	2448 noise_only\11.wav,noise_only
13.wav	2449 noise_only\110.wav,noise_only
14.wav	2450 noise_only\111.wav,noise_only
15.wav	2451 noise_only\112.wav,noise_only
16.wav	2452 noise_only\113.wav,noise_only
17.wav	2453 noise_only\114.wav,noise_only
18.wav	2454 noise_only\115.wav,noise_only
19.wav	2455 noise_only\116.wav,noise_only
20.wav	2456 noise_only\117.wav,noise_only
21.wav	2457 noise_only\118.wav,noise_only
22.wav	2458 noise_only\119.wav,noise_only
23.wav	2459 noise_only\12.wav,noise_only
24.wav	2460 noise_only\120.wav,noise_only
25.wav	2461 noise_only\121.wav,noise_only
26.wav	2462 noise_only\122.wav,noise_only
27.wav	2463 noise_only\123.wav,noise_only
28.wav	2464 noise_only\124.wav,noise_only
29.wav	2465 noise_only\125.wav,noise_only
30.wav	2466 noise_only\126.wav,noise_only
31.wav	2467 noise_only\127.wav,noise_only
32.wav	2468 noise_only\128.wav,noise_only
	2469 noise_only\129.wav,noise_only
	2470 noise_only\13.wav,noise_only
	2471 noise_only\130.wav,noise_only
	2472 noise_only\131.wav,noise_only
	2473 noise_only\132.wav,noise_only
	2474 noise_only\133.wav,noise_only

2.2 Self-Collected Dataset

1. Audio Collection & Pre-processing

- Download via yt-dlp.
- Normalize to 16kHz mono WAV (**ffmpeg**).
- Mute intros/ads (**SponsorBlock**).

2. Source Separation & Diarization

- Isolate vocals (**htdemucs**).
- Generate preliminary timestamps (**pyannote**).

3. Automatic Speech Recognition

- Generate timestamped transcripts (**faster-whisper**).
- Rescan silences to avoid missing speech.

4. Boundary & Segment Refinement

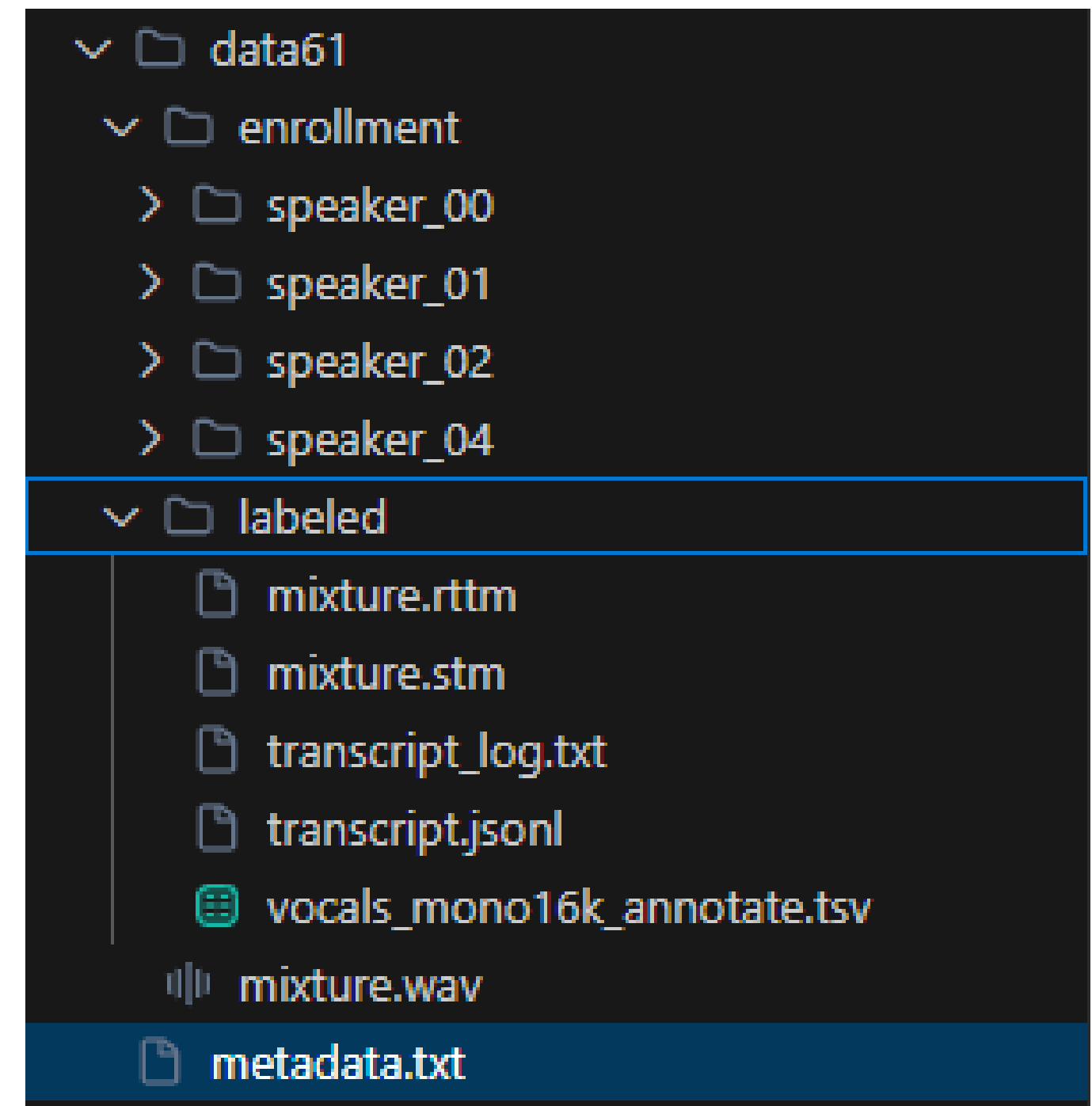
- Load draft into VAD tool.
- Adjust exact start/end boundaries.

5. Transcript & Speaker Correction

- Correct ASR errors.
- Use Auto ASR for difficult segments.

6. Ground-Truth Export

- RTTM, STM/CTM, labels & metadata.



2.2 Dataset Preparation

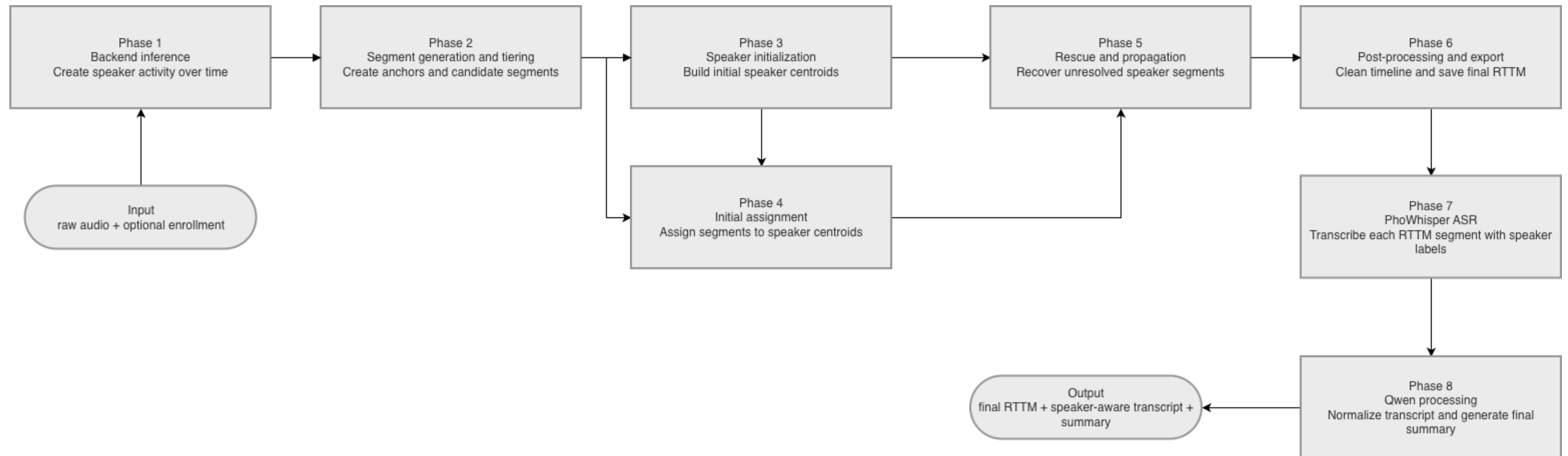
Self-labeled dataset: YouTube

Table 10: YouTube dataset summary for Self-labeled data.

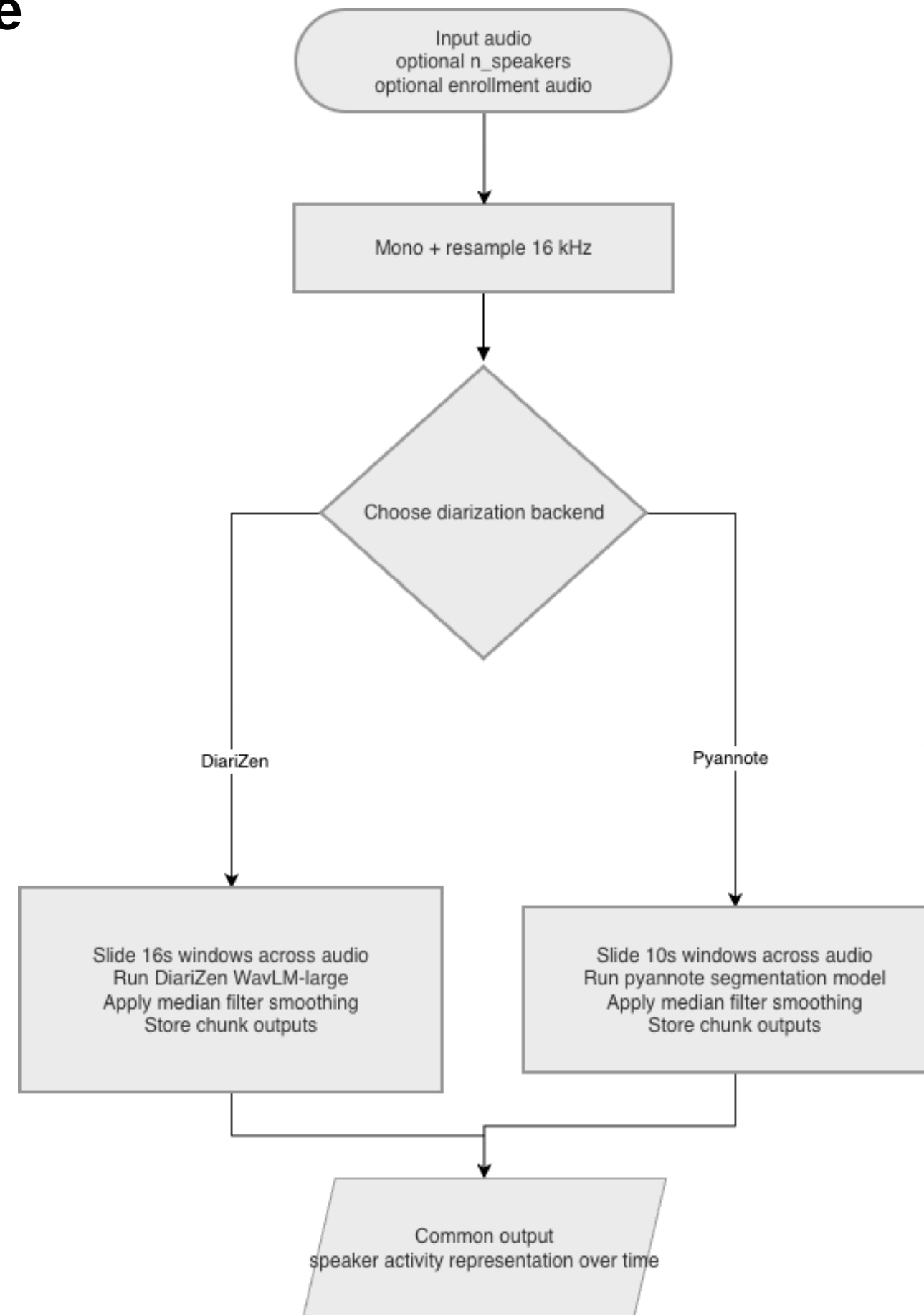
Dataset	Avg. Speakers	Total Hours	Sampling Rate	Usage
Chuyen Ho Chuyen Minh	7	16.97 h	16,000 Hz	Training (Diarization + ASR), Evaluation (Diarization + ASR)
Coi mo	5	6.23 h	16,000 Hz	Training (Diarization + ASR), Evaluation (Diarization + ASR)
Conan	13	11.27 h	16,000 Hz	Training (Diarization + ASR)
Dustin on go	4	2.41 h	16,000 Hz	Training (Diarization + ASR), Evaluation (Diarization + ASR)
VIF	4	9.29 h	16,000 Hz	Training (Diarization + ASR), Evaluation (Diarization + ASR)

3 - System Architecture

3.1 Pipeline Overview



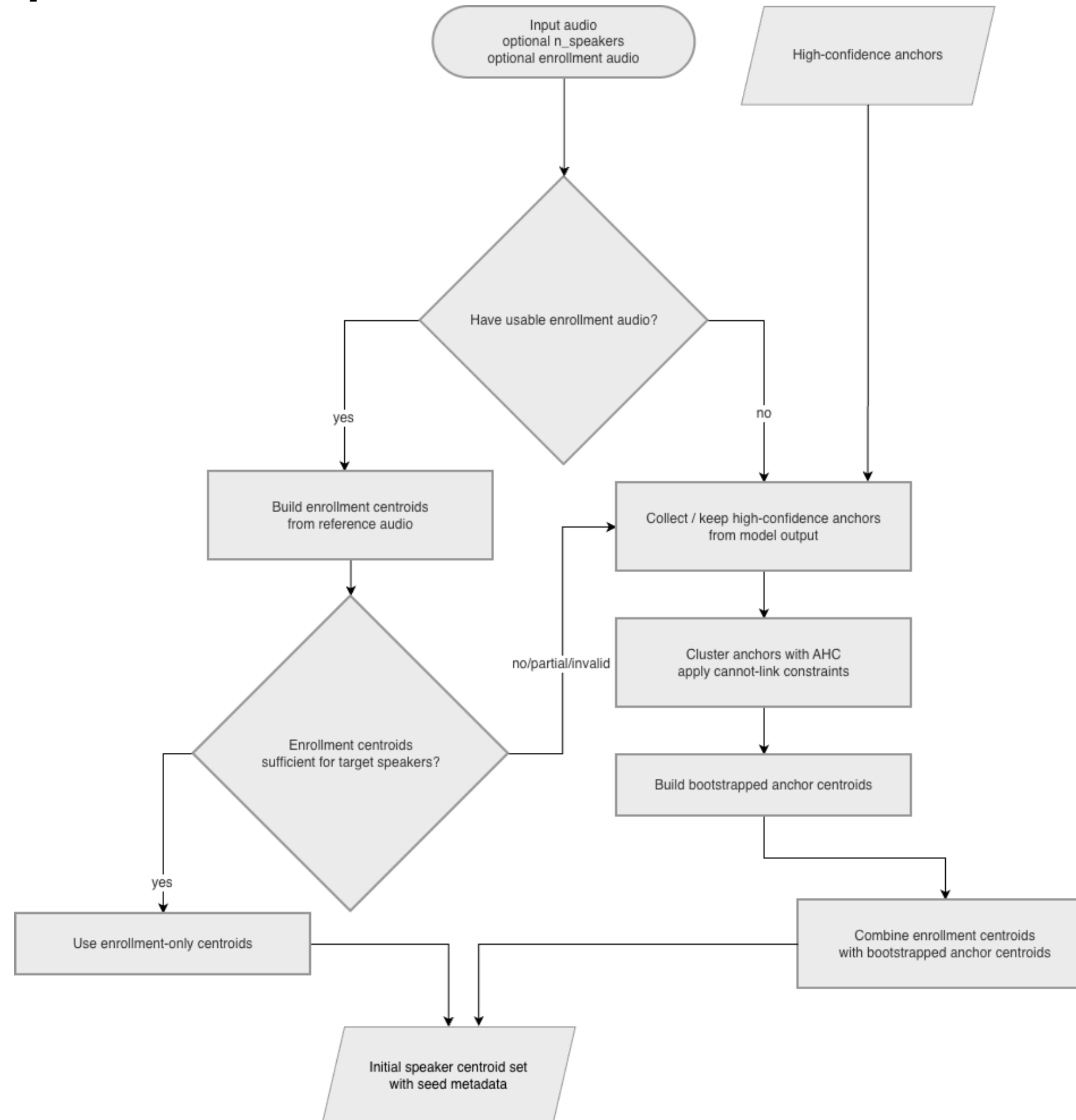
Phase 1: Backend inference



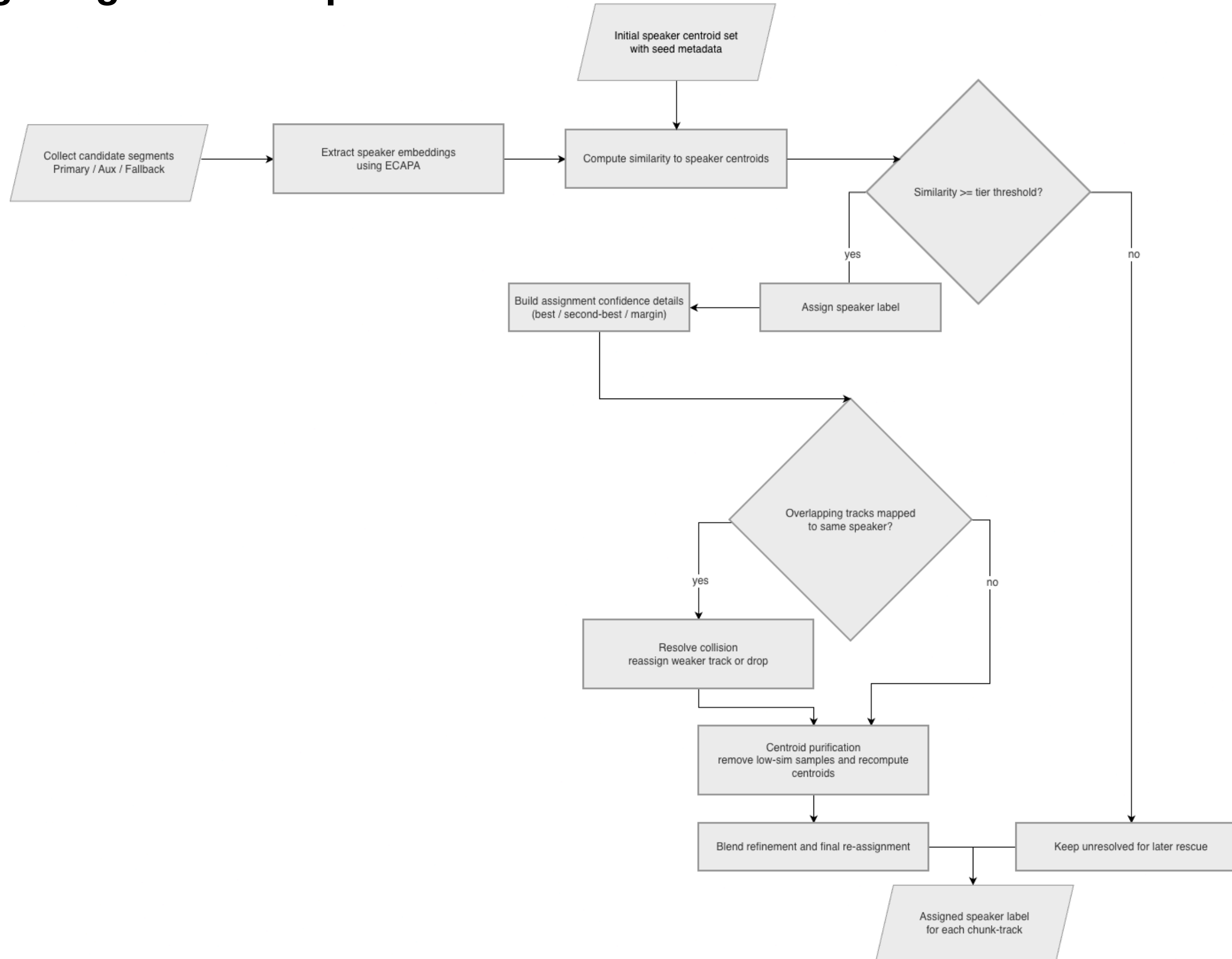
Phase 2: Segment generation and tiering Create anchors



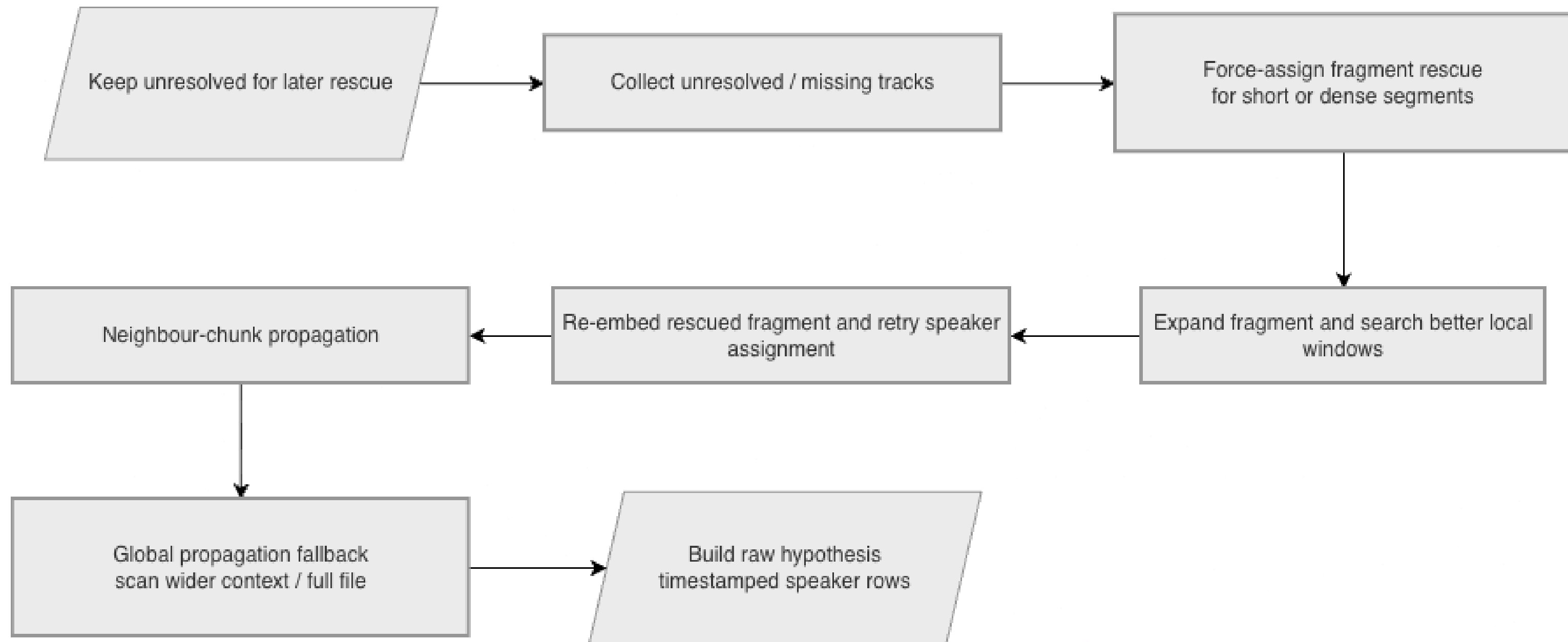
Phase 3: Build initial speaker centroids



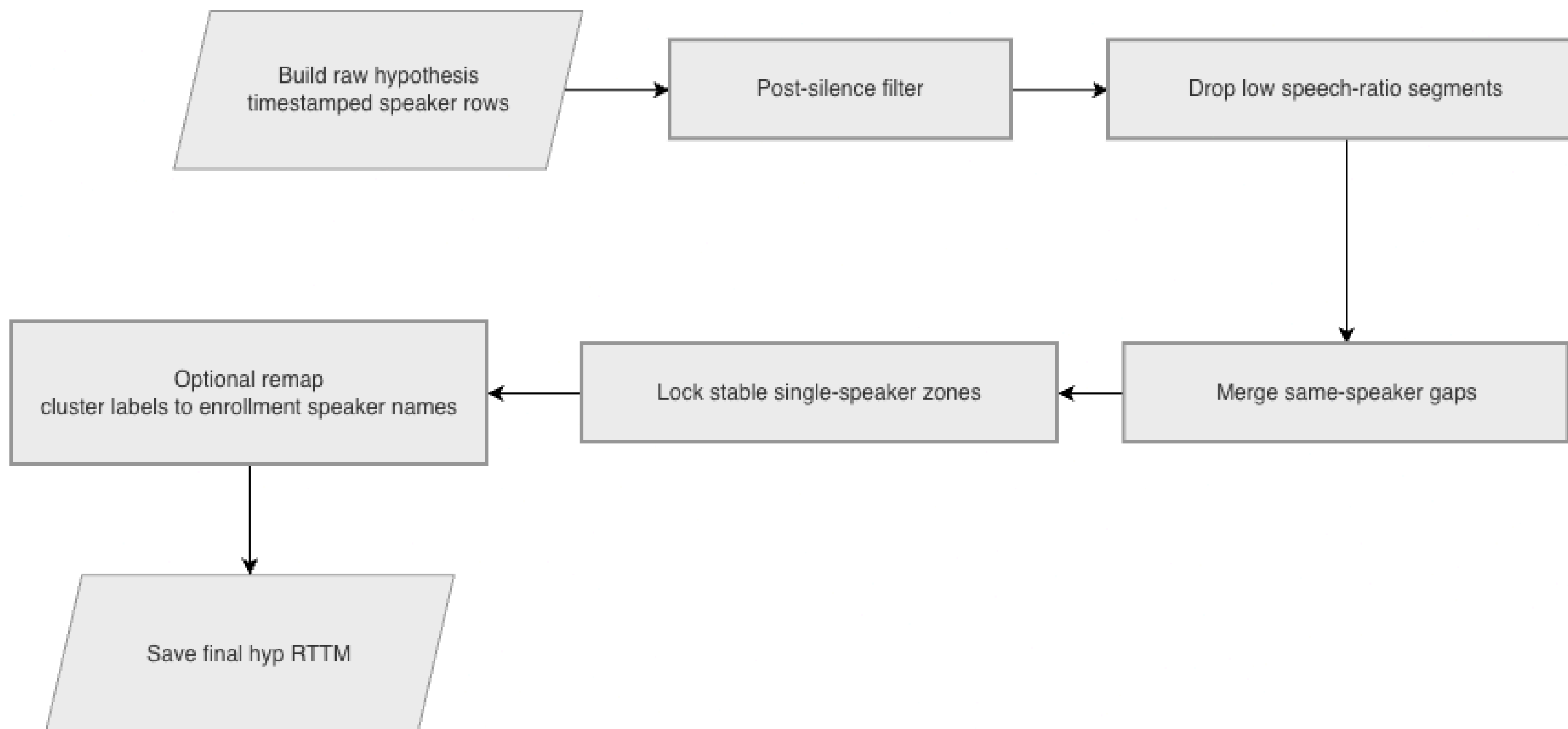
Phase 4: Assign segments to speaker centroids



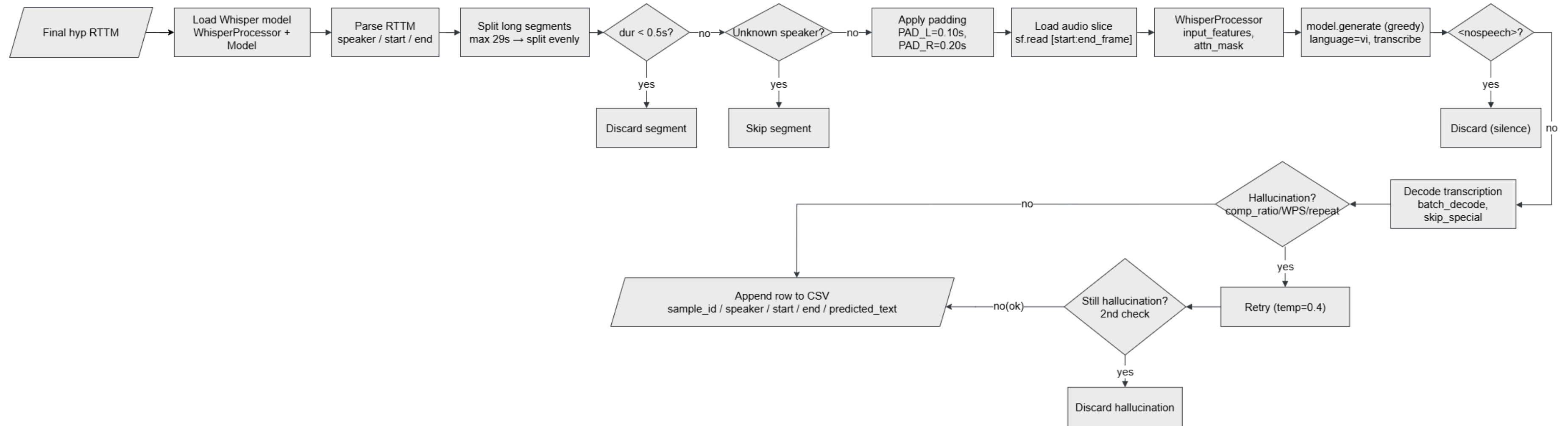
Phase 5: Rescue and propagation



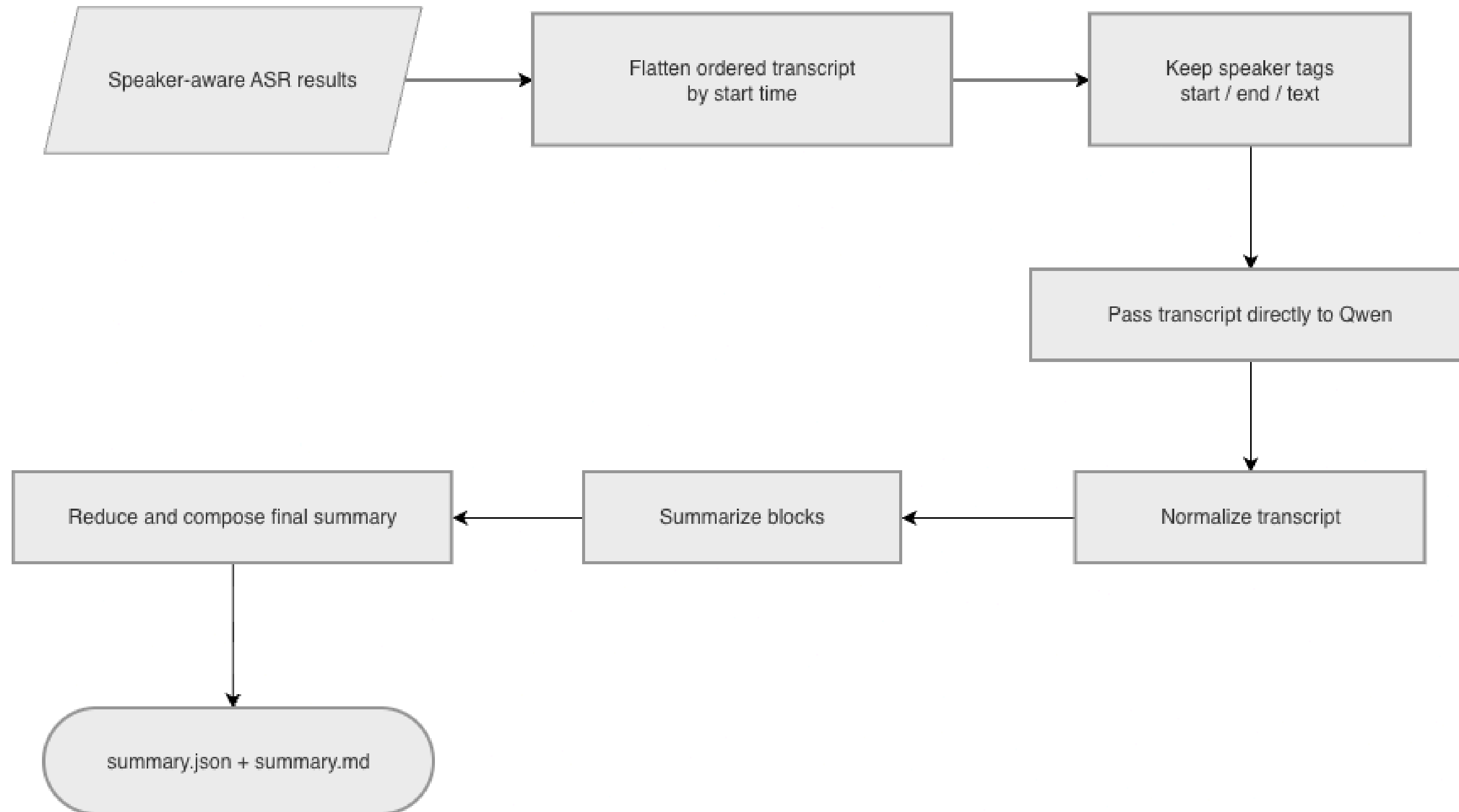
Phase 6: Post-processing



Phase 7: PhoWhisper ASR

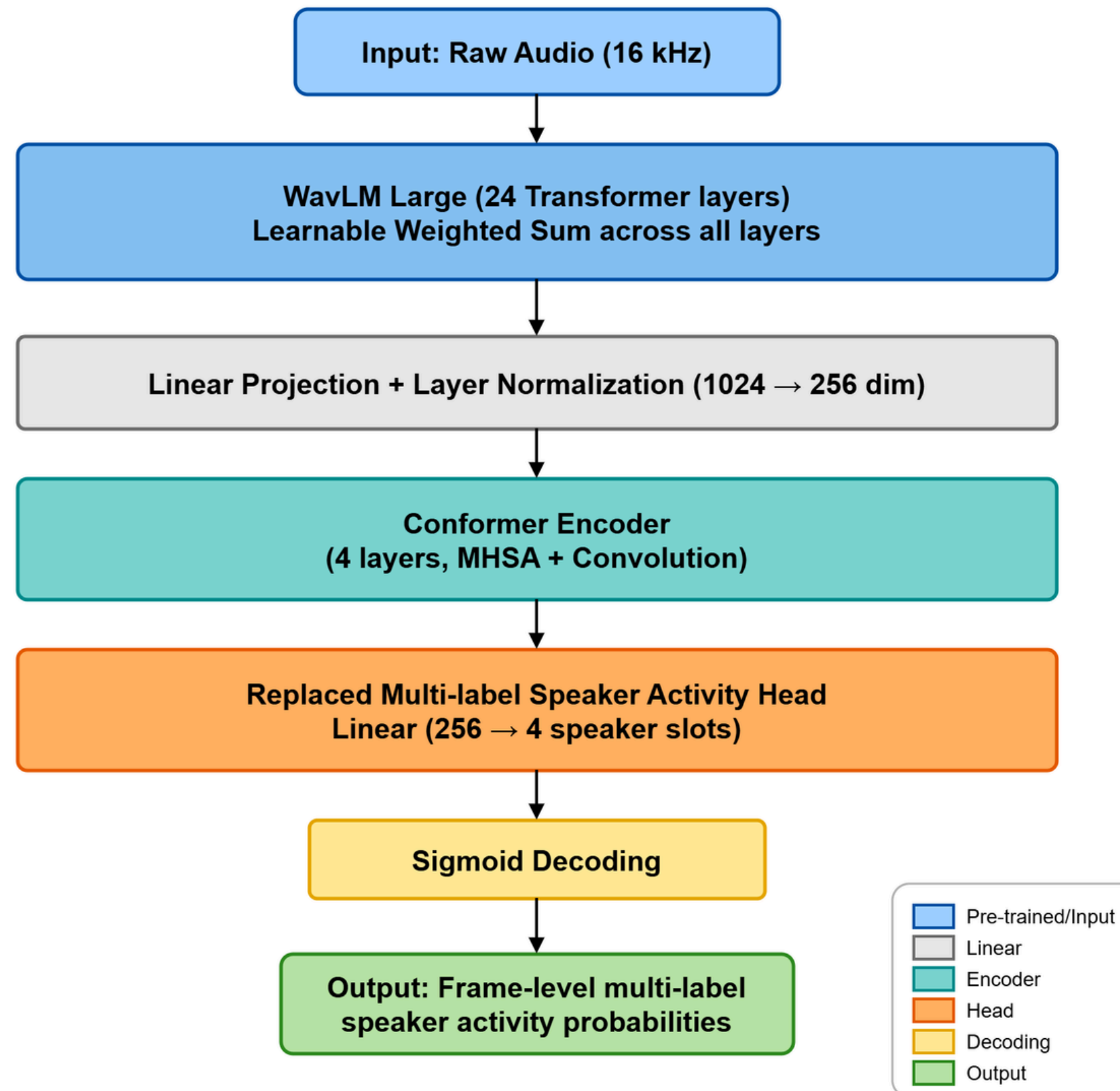


Phase 8: Qwen processing

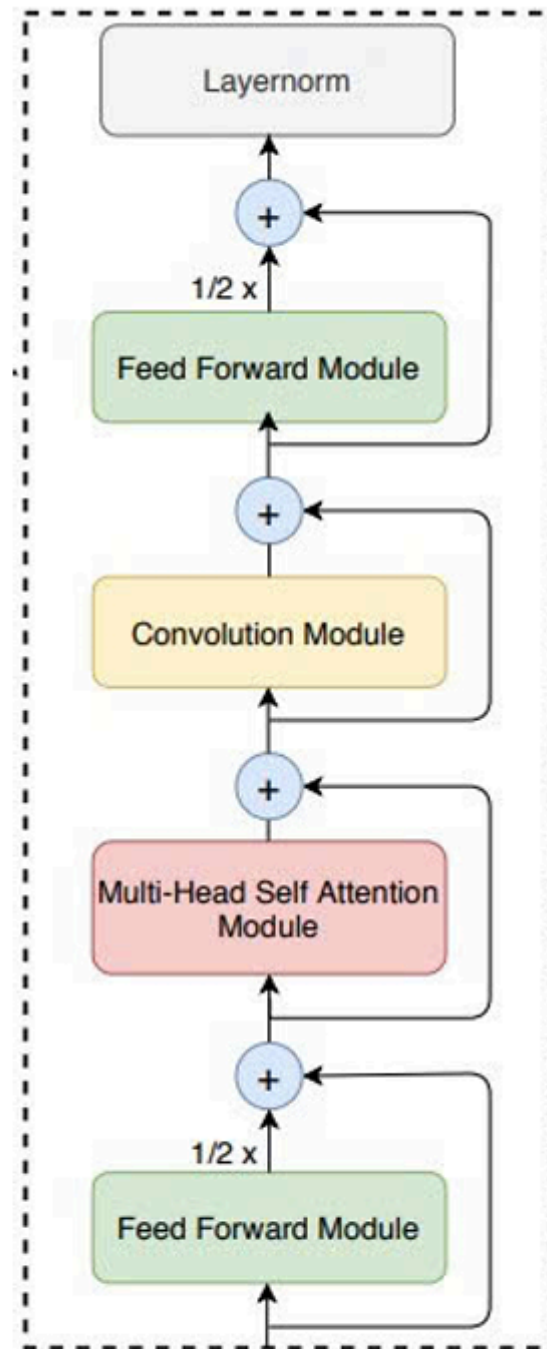


3.2 DiariZen Model — Overview and Architecture

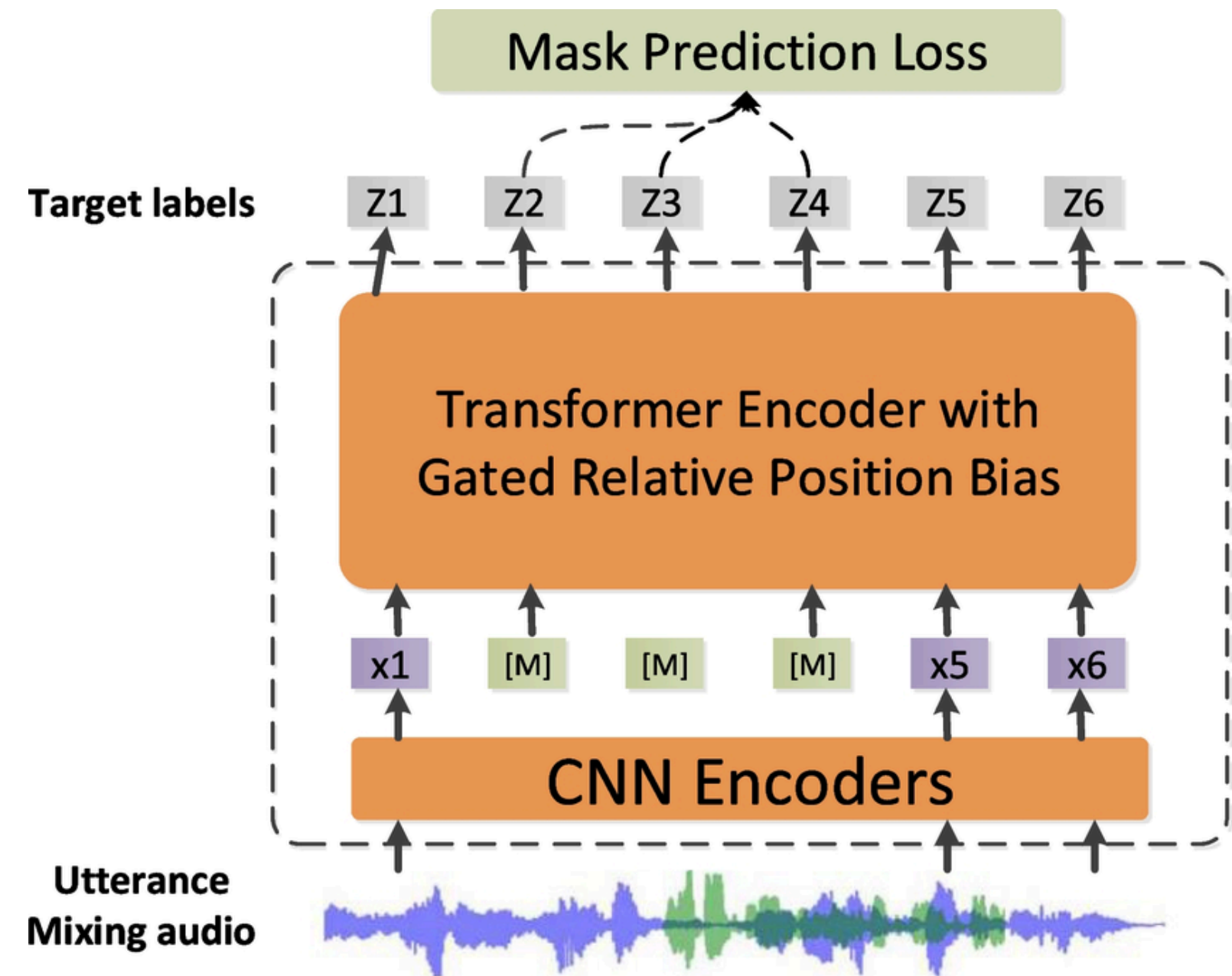
Modified EEND Module Architecture in DiariZen



3.2 DiariZen Model — Overview and Architecture



Conformer block



Wavlm

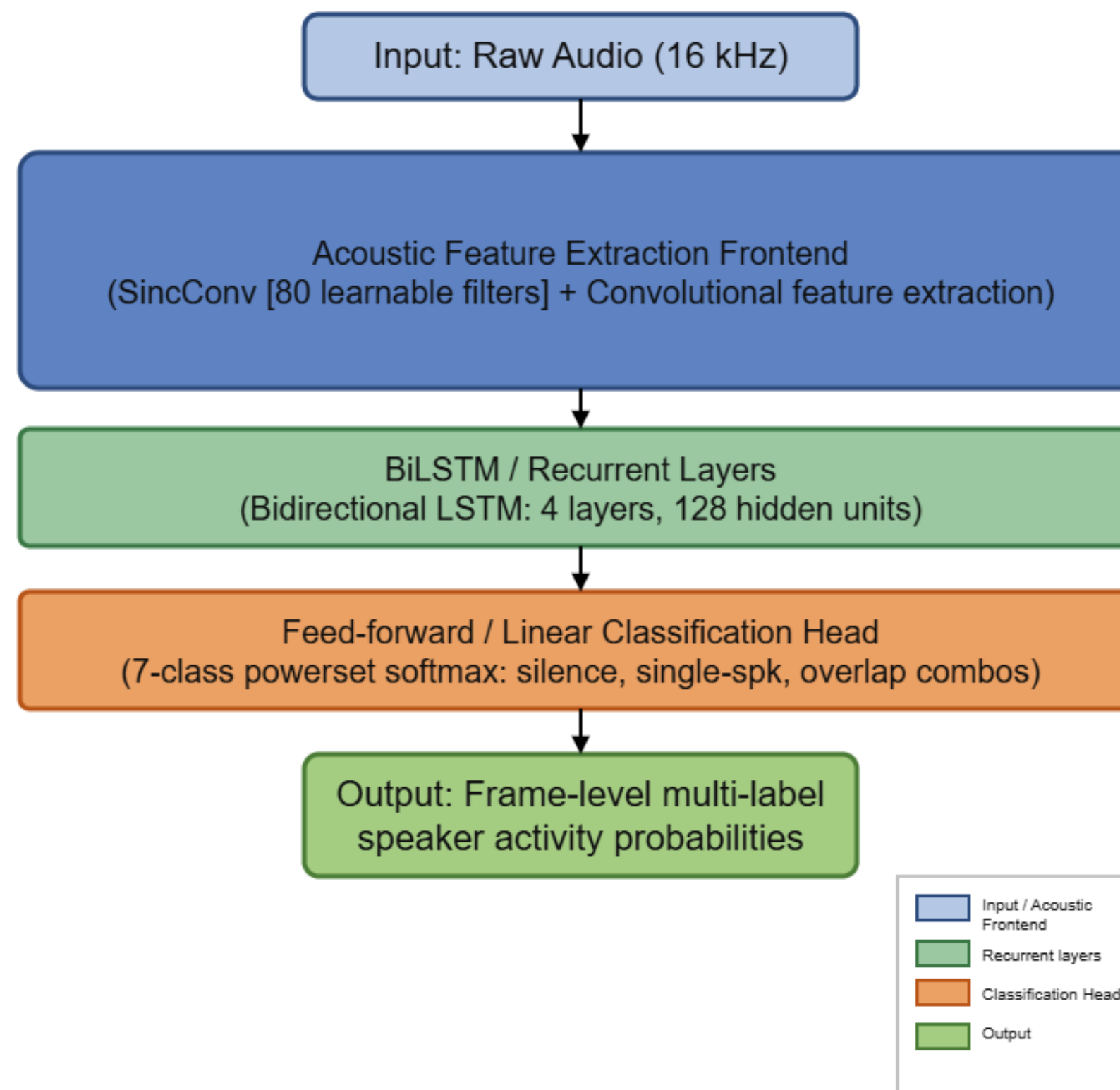
3.2 DiariZen Model — PIT-BCE Loss

$$\mathcal{L}_{\text{PIT}} = \min_{\sigma \in \text{Perm}(K)} \frac{1}{T \cdot K} \sum_{t=1}^T \sum_{k=1}^K \text{BCE}(\hat{y}_{t,k}, y_{t,\sigma(k)})$$

K = speaker slots | σ^* = best permutation | Gradient only through σ^* | BCE = Binary Cross-Entropy

3.3 Pyannote Pipeline — Overview and Architecture

Modified EEND Module Architecture in Pyannote



	y_1													y_t	y_T			
$\emptyset$	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0
$\{s_1\}$	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$\{s_2\}$	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0
$\{s_3\}$	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1
$\{s_1, s_2\}$	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
$\{s_1, s_3\}$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

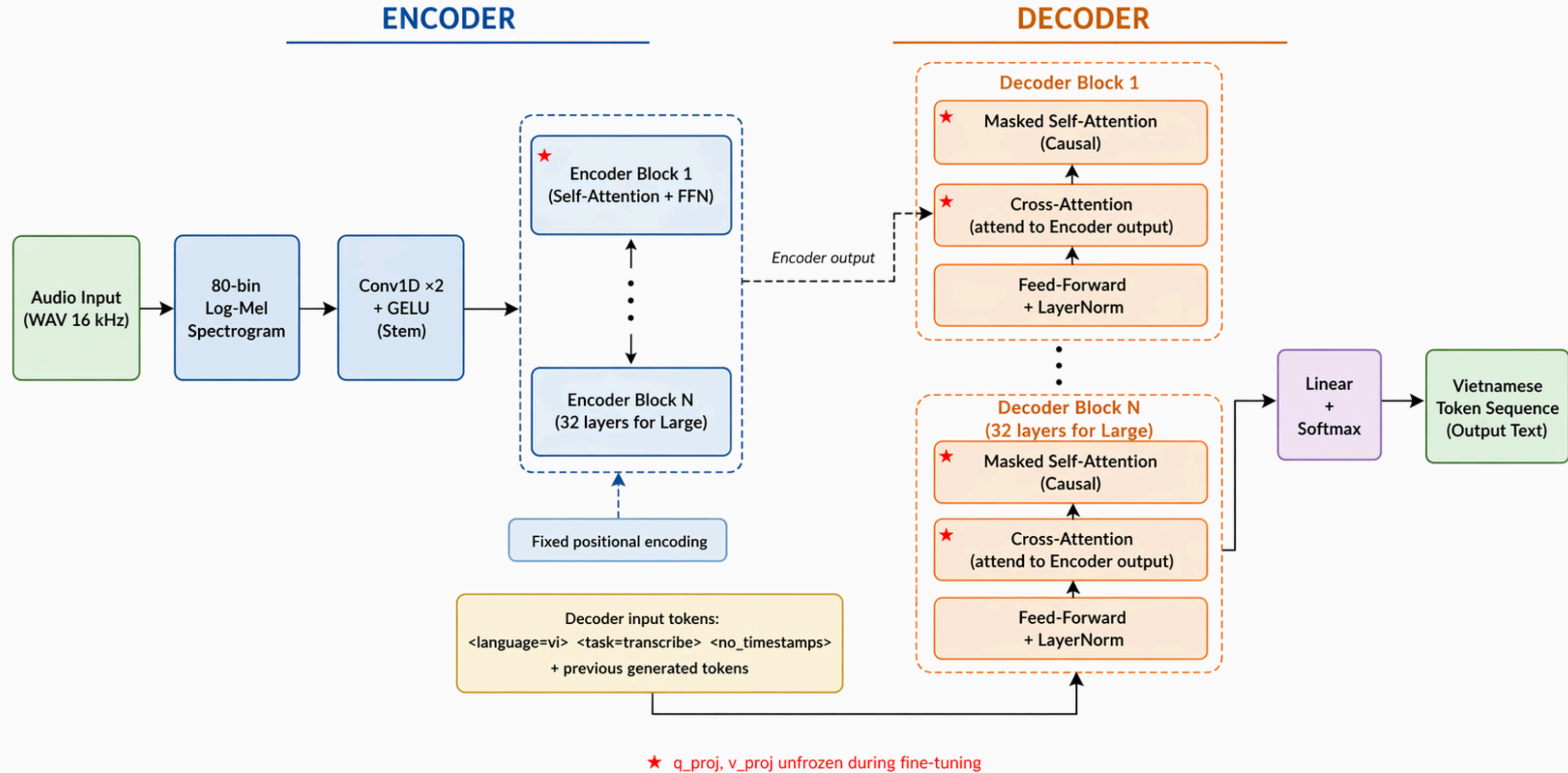
3.3 Pyannote — Powerset CE + PIT

$$\mathcal{L}_{\text{PS}} = - \sum_{t=1}^T \sum_{c=1}^C y_{t,c} \log \hat{p}_{t,c} \quad C = 1 + K + \frac{K(K-1)}{2}$$

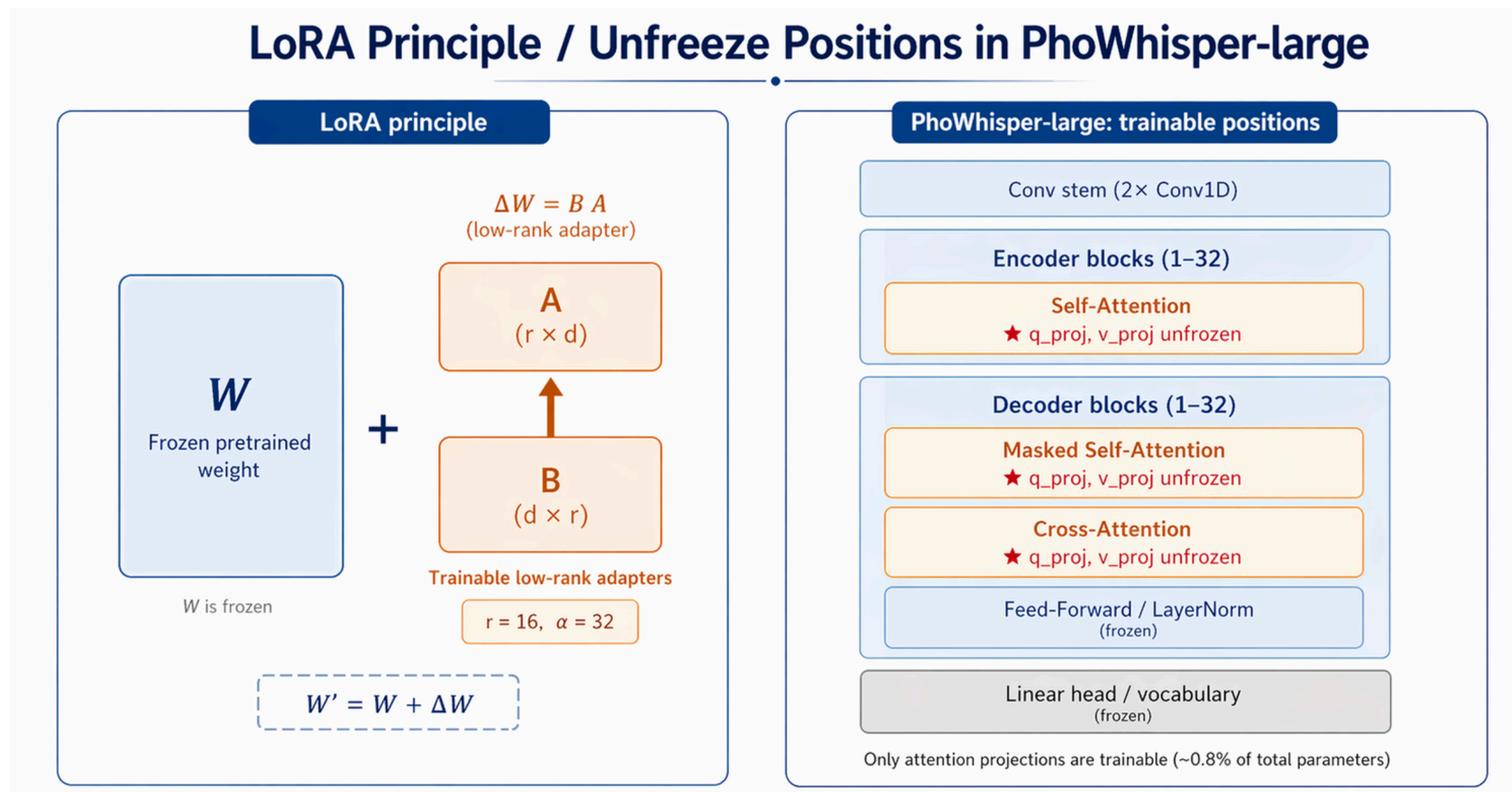
C = powerset classes (e.g. $K = 3 \Rightarrow C = 7$) | Decision: $\arg\max_c \hat{p}_{t,c}$ | No external threshold needed

3.4 PhoWhisper-large — Overview and Architecture

PhoWhisper-large Encoder-Decoder Architecture



3.4 PhoWhisper-large — LoRA and Trainable Positions

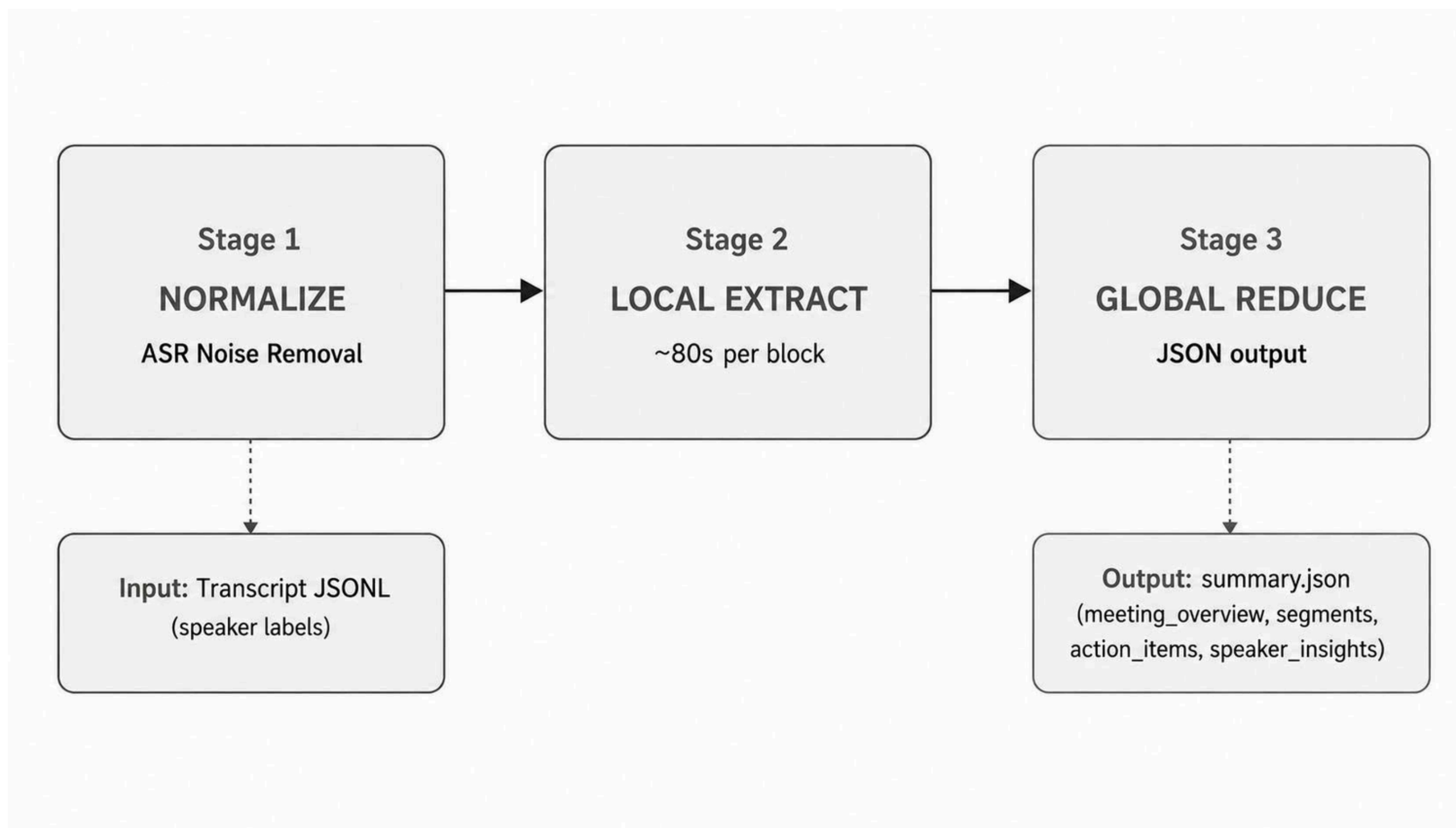


3.4 PhoWhisper-large — Loss

$$\mathcal{L}_{\text{CE}} = -\frac{1}{T} \sum_{t=1}^T \log P_{\theta}(y_t \mid y_{<t}, \mathbf{x})$$

T = sequence length | y_t = target token at step t | $\mathbf{x}$ = audio input | θ = model parameters

3.5 Qwen Summarization Pipeline

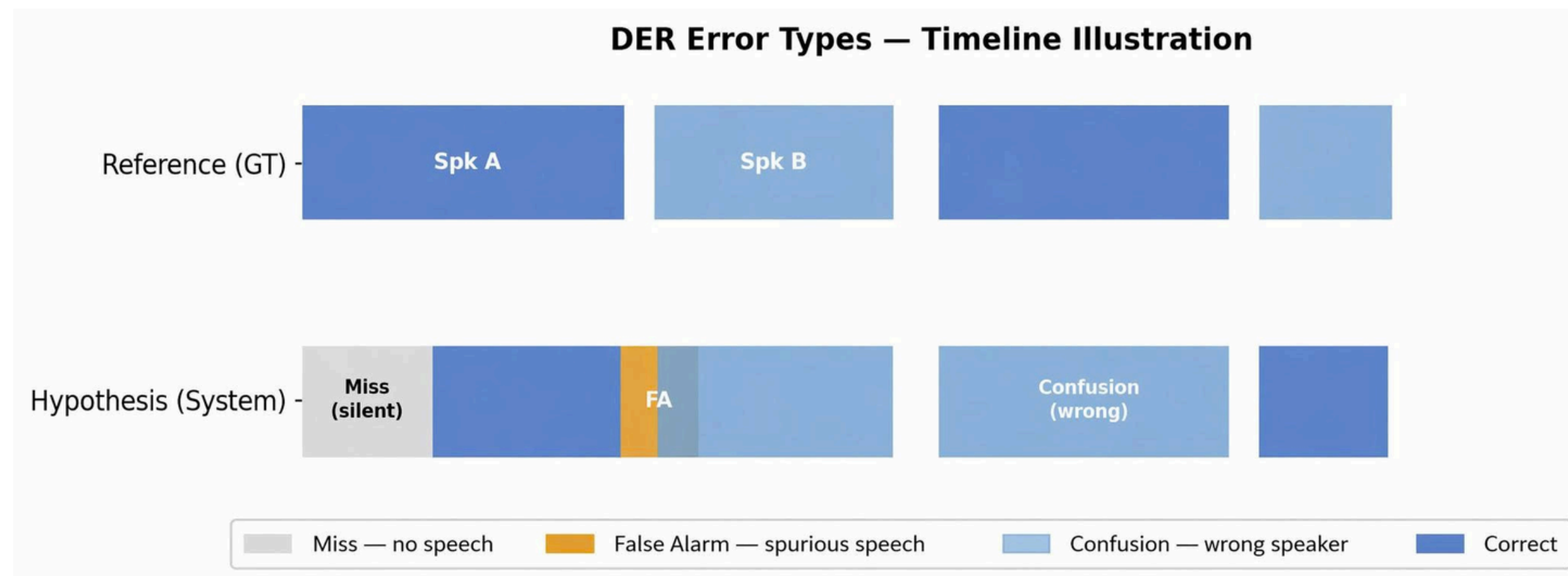


4 - Evaluation Metrics

4.1 Diarization Error Rate (DER)

$$\text{DER} = \frac{T_{\text{miss}} + T_{\text{fa}} + T_{\text{conf}}}{T_{\text{ref}}} \times 100\%$$

T_{ref} = total reference speech duration | T_{miss} = missed speech | T_{fa} = false alarm | T_{conf} = speaker confusion



4.2 DER Components: False Alarm, Miss, and Confusion

$$\text{FA}\% = \frac{\sum_r \text{dur}(r) (N_{\text{hyp}} - N_{\text{ref}})^+}{\sum_r \text{dur}(r) N_{\text{ref}}} \times 100\%$$

N_{ref} = number of reference speakers in region r | N_{hyp} = number of hypothesis speakers

$$\text{Miss}\% = \frac{\sum_r \text{dur}(r) (N_{\text{ref}} - N_{\text{hyp}})^+}{\sum_r \text{dur}(r) N_{\text{ref}}} \times 100\%$$

N_{ref} = number of reference speakers in region r | N_{hyp} = number of hypothesis speakers

$$\text{Conf}\% = \frac{\sum_r \text{dur}(r) (\min(N_{\text{hyp}}, N_{\text{ref}}) - N_{\text{cor}})}{\sum_r \text{dur}(r) N_{\text{ref}}} \times 100\%$$

N_{cor} = number of correctly matched speakers after optimal Hungarian assignment | N_{ref} = number of reference speakers in region r | N_{hyp} = number of hypothesis speakers

4.3 Overlap Metrics and Word Error Rate

$$\text{Precision} = \frac{|O_{\text{hyp}} \cap O_{\text{ref}}|}{|O_{\text{hyp}}|} \quad \text{Recall} = \frac{|O_{\text{hyp}} \cap O_{\text{ref}}|}{|O_{\text{ref}}|} \quad \text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$|\cdot|$ = total duration (s) | O_{hyp} = predicted overlap | O_{ref} = actual overlap

$$\text{WER} = \frac{S + D + I}{N} \times 100\%$$

S = substitutions | D = deletions | I = insertions | N = total reference words

4.4 Main Metrics by System Component

System Component	Main Metrics
Diarization	DER ↓, FA ↓, Miss ↓, Conf ↓, P ↑, R ↑ F1↑
ASR	WER ↓
Speaker Identification	Confusion ↓

4.5 Automated Summary Evaluation and LLM Judge Criteria

Table 42: Automated Evaluation Metrics

Metric	Good when	Definition
Overall Score	High (max 1.0)	Aggregate score produced by the evaluation script, combining schema validity, evidence support, speaker-specificity, and confidence quality. Higher is better.
Schema Pass Rate	= 1.0	Rate of JSON outputs with all required keys present
Evidence Support Rate	= 1.0	Rate of speaker-summary items whose evidence spans are valid and grounded in the transcript timeline.
Speaker Presence Rate	= 1.0	Rate of expected speakers that are mentioned in the output.
Generic Speaker Summary Rate	= 0.0	Rate of generic, non-specific speaker summaries. Lower is better.
Low Confidence Rate	= 0.0	Rate of speaker summaries marked with low confidence. Lower is better.

Table 43: LLM Judge Criteria (Qwen 7B)

Criterion	Max Score	Definition
Main Summary Score	4	Accuracy of the main conversation summary
Speaker Summary Score	4	Accuracy of per-speaker summaries
Faithfulness Score	2	Information is not fabricated; grounded in the original transcript
Conciseness Score	2	Summary is concise, not verbose or repetitive

5. Training and Fine-tuning

5.1 Setup

Training

- **Operating System:** Windows 10 (64-bit)
- **GPU:** NVIDIA RTX 3090 Ti 24GB
- **CUDA Drivers:** 12.6
- **RAM:** 88 GB
- **Storage:** 1500 GB
- **Python:** 3.10.11
- **Development Environment:** VS Code

Demo

Operating System: Windows 11 (64-bit)
GPU: NVIDIA RTX 4060 8 GB
CUDA Drivers: 12.8
RAM: 16 GB
Storage: 512 GB
Python: 3.10.11
Development Environment: VS Code

5.2 Fine-tuning Data for Diarization - Idea finetune

Idea	Reason
Keep the pretrained backbone	The backbone has already learned strong speech representations, so it does not need to be trained from scratch.
Replace or adjust the head	The output format or the number of speaker slots in the new task is different from the original model.
Gradual unfreezing	This helps avoid catastrophic forgetting by gradually unfreezing higher layers.
Higher LR for the head than the backbone	The new head needs to learn quickly, while the backbone only needs light adaptation.
Curriculum data mix	Training starts with easier and more stable data, then gradually increases overlap and noise.
PIT loss	This solves the speaker-slot permutation problem.
ECAPA embedding + clustering	This connects local speakers across the whole recording and supports files with more global speakers than the local speaker-slot limit.

5.2 Fine-tuning Data for Diarization

Table 26: Self-labeled data distribution.

Data Source	Set	Files	Total Hours	Speaker Range
TRAIN				
Chuyen_Ho_Chuyen_Minh	Train	43	14.77 h	4-13
VIF	Train	20	3.37 h	3-6
Coi_mo	Train	15	4.22 h	4-5
Dustin_on_go	Train	8	1.74 h	4
Total Train self-labeled		86	24.09 h	3-13
VALIDATION				
VIF	Val	2	2.15 h	4-5
Dustin_on_go	Val	1	1.19 h	4
Chuyen_Ho_Chuyen_Minh	Val	1	0.99 h	4
Coi_mo	Val	2	0.94 h	4-5
Total Val self-labeled		6	5.27 h	4-5
TEST				
VIF	Test	2	2.52 h	4-5
Chuyen_Ho_Chuyen_Minh	Test	1	1.21 h	4
Dustin_on_go	Test	1	1.19 h	4
Coi_mo	Test	2	1.07 h	4-5
Total Test self-labeled		6	6.00 h	4-5
TOTAL SELF-LABELED		98	35.36 h	3-13

Table 27: Synthetic Data distribution data_mix.

Data Group	Set	Files	Total Hours	Speaker Range
TRAIN				
0% overlap	Train	43	26.8 h	2-4
5% overlap	Train	39	24.4 h	2-4
10% overlap	Train	39	24.3 h	2-4
Total Synthetic Data		121	75.54 h	2-4
VALIDATION				
Synthetic data	Val	10	4.66 h	2-4
Total Val Synthetic Data		10	4.66 h	2-4
TEST				
Synthetic data	Test	10	4.70 h	2-4
Total Test Synthetic Data		10	4.70 h	2-4
TOTAL Synthetic Data		141	84.90 h	2-4

5.3 Fine-tuning Parameters — DiariZen (BUT-FIT)

Table 28: Detailed fine-tuning configuration: DiariZen (BUT-FIT/diarizen-wavlm-large-s80-md-v2)

Parameter	Value
Model & Architecture	
Base Model	BUT-FIT/diarizen-wavlm-large-s80-md-v2
Backbone	WavLM Large (24 layers, 316M params)
Decoder	Conformer + Classification Head
Head (after replacement)	Linear(out=4) + Sigmoid (multilabel, 4 speakers)
Head Init	Xavier uniform
Fine-tuning Strategy: Gradual Unfreezing	
Freeze strategy	top_N (progressively unfreezing from top)
Phase 1: Unfreeze	top-4 layers
Phase 1: LR	5e-5
Phase 1: Epochs	4
Phase 1: Augmentation	False
Phase 1: Data mix	80% self-labeled; 15% tier B (overlap 5-10%); 5% tier C (overlap 15%)
Phase 2: Unfreeze	top-6 layers
Phase 2: LR	2e-5
Phase 2: Epochs	6
Phase 2: Augmentation	True
Phase 2: Data mix	60% self-labeled; 30% tier B; 10% tier C
Phase 3: Unfreeze	top-6 layers
Phase 3: LR	1e-5
Phase 3: Epochs	4
Phase 3: Augmentation	True
Phase 3: Data mix	55% self-labeled; 25% tier B; 20% tier C

Loss Function & Weights	
Loss function	PIT multilabel BCE (pit_loss)
overlap_loss_weight	1.7 (phase 2-3); 1.4 (phase 1)
silence_loss_weight	1.0
triple_loss_weight	0.0: triple overlap frames masked from gradient
boundary_ignore_sec	downweight × 0.25 for boundary frames
Optimization	
Batch Size	6
Gradient Accumulation	2 (effective batch = 12)
Optimizer	AdamW
Gradient Clipping	0.5
Weight Decay	0.01
Precision	Mixed Precision BFloat16 (bfloat16, AMP)
Inference Pipeline Configuration	
Chunk size	16.0 s (CHUNK_SEC = 16.0)
Sampling rate	16 kHz
Decode threshold	0.25 (sigmoid multilabel)
Median filter window	MED_WIN = 11 frames
High centroid threshold	HIGH_THR_FIXED = 0.70
Default n_clusters	FIXED_N_CLUSTERS = 3
Max speakers	4
Validation	
Crop-level validation	BCE-PIT loss + PIT-DER on 16 s crops
File-level validation	DER with collar = 0.25 s (pyannote.metrics)
Metric for best ckpt	file-level DER (lower is better)

5.4 Fine-tuning Parameters — Pyannote

Parameter	Value
Model & Architecture	
Base Pipeline	pyannote/speaker-diarization-community-1
Segmentation model	pyannote/segmentation-3.0
Embedding model	speechbrain/spkrec-ecapa-voxceleb
Clustering	Agglomerative Hierarchical Clustering (AHC)
model_max_speakers_per_chunk	3 (CHECK-ONLY, policy = warn)
model_max_speakers_per_frame	2
max_speakers (file level)	4
eval_max_total_speakers	4
Fine-tuning Strategy: Gradual Unfreezing	
Freeze strategy	top_N (progressively unfreezing from top)
Phase 1: Unfreeze	top-4 layers (phase1_top4_anchor)
Phase 1: LR	5e-5
Phase 1: Epochs	4
Phase 1: Augmentation	False
Phase 1: Data mix	80% self-labeled 15% tier B 5% tier C
Phase 2: Unfreeze	top-6 layers (phase2_top6_overlap)
Phase 2: LR	2e-5
Phase 2: Epochs	6
Phase 2: Augmentation	True
Phase 2: Data mix	60% self-labeled 30% tier B 10% tier C
Phase 3: Unfreeze	top-6 layers (phase3_top6 Consolidate)
Phase 3: LR	1e-5
Phase 3: Epochs	4
Phase 3: Augmentation	True
Phase 3: Data mix	55% self-labeled 25% tier B 20% tier C

Loss Function & Weights	
Loss function	Powerset cross-entropy + PIT
overlap_loss_weight	1.7 (phase 2-3); 1.4 (phase 1)
silence_loss_weight	1.0
Decode mode (validation)	argmax_powerset (no external threshold used)
Optimization	
Batch Size	128
Gradient Accumulation	1
Optimizer	AdamW
Gradient Clipping	0.5
Weight Decay	0.01
Precision	Mixed Precision BFloat16 (bfloat16, AMP)
Inference Pipeline Configuration	
Sampling rate	16 kHz
Decode mode	argmax on powerset classes (no external threshold)
Threshold sweep	Almost no effect due to argmax-based decode
Validation	
Crop-level validation	Powerset cross-entropy + PIT-DER on short crops
File-level validation	DER with collar = 0.25s (pyannote.metrics)
Metric for best ckpt	file-level DER (lower is better)
Training history	Saved to training_history.json

5.6 Fine-tuning Parameters — PhoWhisper

Fine-tuning Strategy	Trainable Scope	Trainable Parameters	Main Advantage	Main Limitation
Full Fine-tuning	All PhoWhisper encoder-decoder layers	100%	Highest adaptation capacity to domain-specific speech data	High VRAM cost, longer training time, higher overfitting risk
Partial Unfreezing	Only q_proj and v_proj attention projection layers	Low	Reduces training cost while adapting key attention components	Still directly modifies original model weights
LoRA on q_proj, v_proj	Frozen PhoWhisper backbone + LoRA adapters on q_proj, v_proj	~0.8%	Best trade-off between adaptation quality, VRAM efficiency, and deployment stability	Depends on LoRA rank, alpha, and adapter configuration

5.5 Fine-tuning Data for PhoWhisper

Table 32: Data Distribution by Domain and Quality Bucket.

Domain	Characteristics	Bucket	Train sets	Training hours
Conan	Anime dubbing, clear voice, low noise	silver_real	25	10,43 h
Chuyen Ho	Natural conversation, fast-paced speech	silver_real	15	14,76 h
Coi Mo	Podcast, heavy overlap, slang vocabulary	hard_real	5	4,33 h
VIF	Standard broadcast television narration	gold_real	3	7,45 h
Dustin On Go	Outdoor vlog, ambient background noise	silver_real	1	1,21 h
TOTAL			49	38,2 h

Table 34: Synthetic Data distribution.

Synthetic Data: TRAIN				
0% overlap	Train	43	26.8 h	2-4 speakers
5% overlap	Train	39	24.4 h	2-4 speakers
10% overlap	Train	39	24.3 h	2-4 speakers
Total Train data_mix		121	75.54 h	2-4 speakers
Synthetic Data: VALIDATION				
Total Val Synthetic Data		10	4.66 h	2-4 speakers
Synthetic Data: TEST				
Total Synthetic Data		10	4.70 h	2-4 speakers
TOTAL Synthetic Data		141	84.90 h	2-4 speakers

Table 33: Self-labeled data distribution.

Data Source	Set	Files	Total Hours	Speaker Range
TRAIN				
Chuyen_Ho_Chuyen_Minh	Train	43	14.77 h	4-13
VIF	Train	20	3.37 h	3-6
Coi_mo	Train	15	4.22 h	4-5
Dustin_on_go	Train	8	1.74 h	4
Total Train self-labeled	Train	86	24.09 h	3-13
VALIDATION				
VIF	Val	2	2.15 h	4-5
Dustin_on_go	Val	1	1.19 h	4
Chuyen_Ho_Chuyen_Minh	Val	1	0.99 h	4
Coi_mo	Val	2	0.94 h	4-5
Total Val self-labeled	Val	6	5.27 h	4-5
TEST				
VIF	Test	2	2.52 h	4-5
Chuyen_Ho_Chuyen_Minh	Test	1	1.21 h	4
Dustin_on_go	Test	1	1.19 h	4
Coi_mo	Test	2	1.07 h	4-5
Total Test self-labeled	Test	6	6.00 h	4-5
TOTAL SELF-LABELED		98	35.36 h	3-13

5.6 Fine-tuning Parameters — PhoWhisper

Table 35: Detailed fine-tuning configuration for PhoWhisper-large.

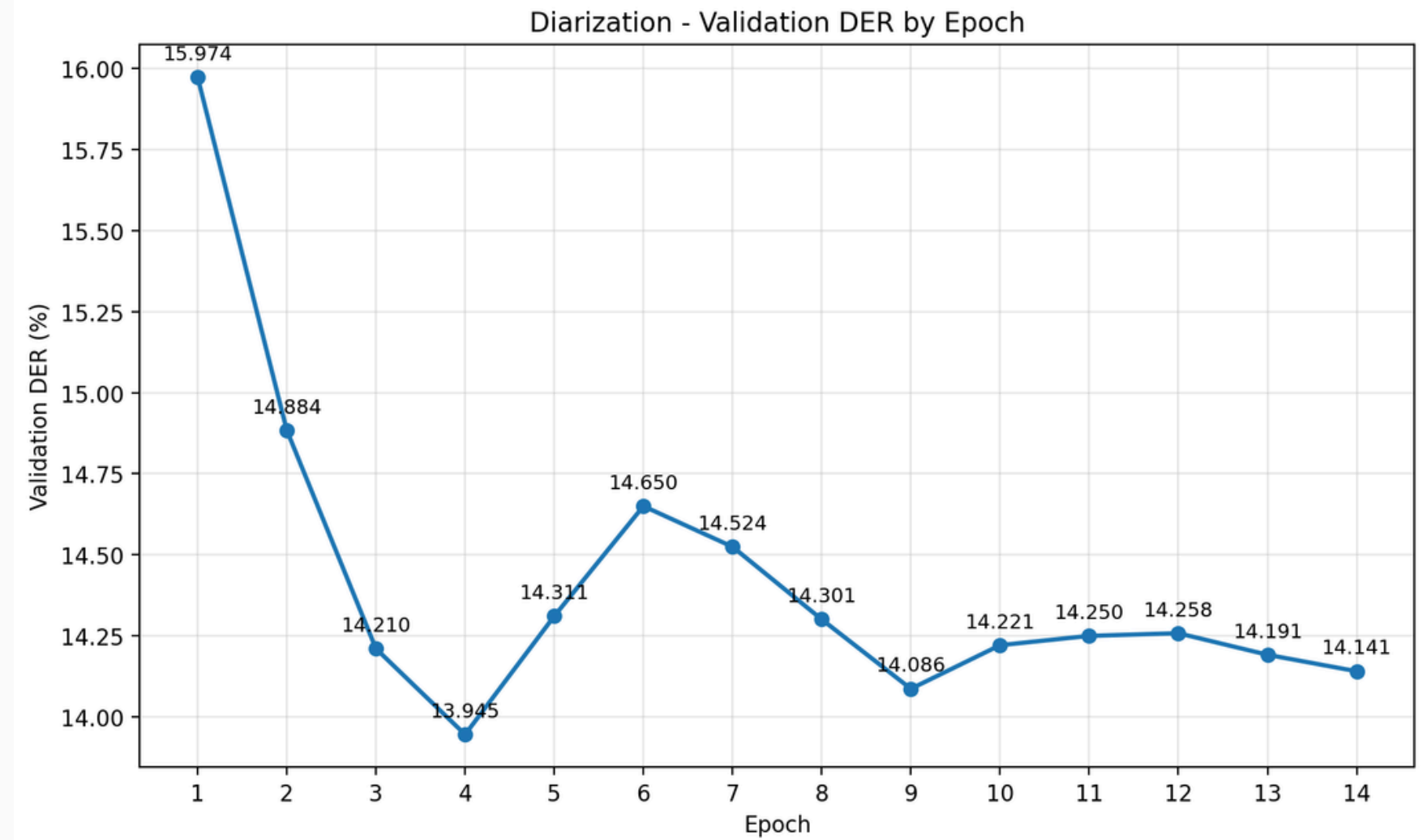
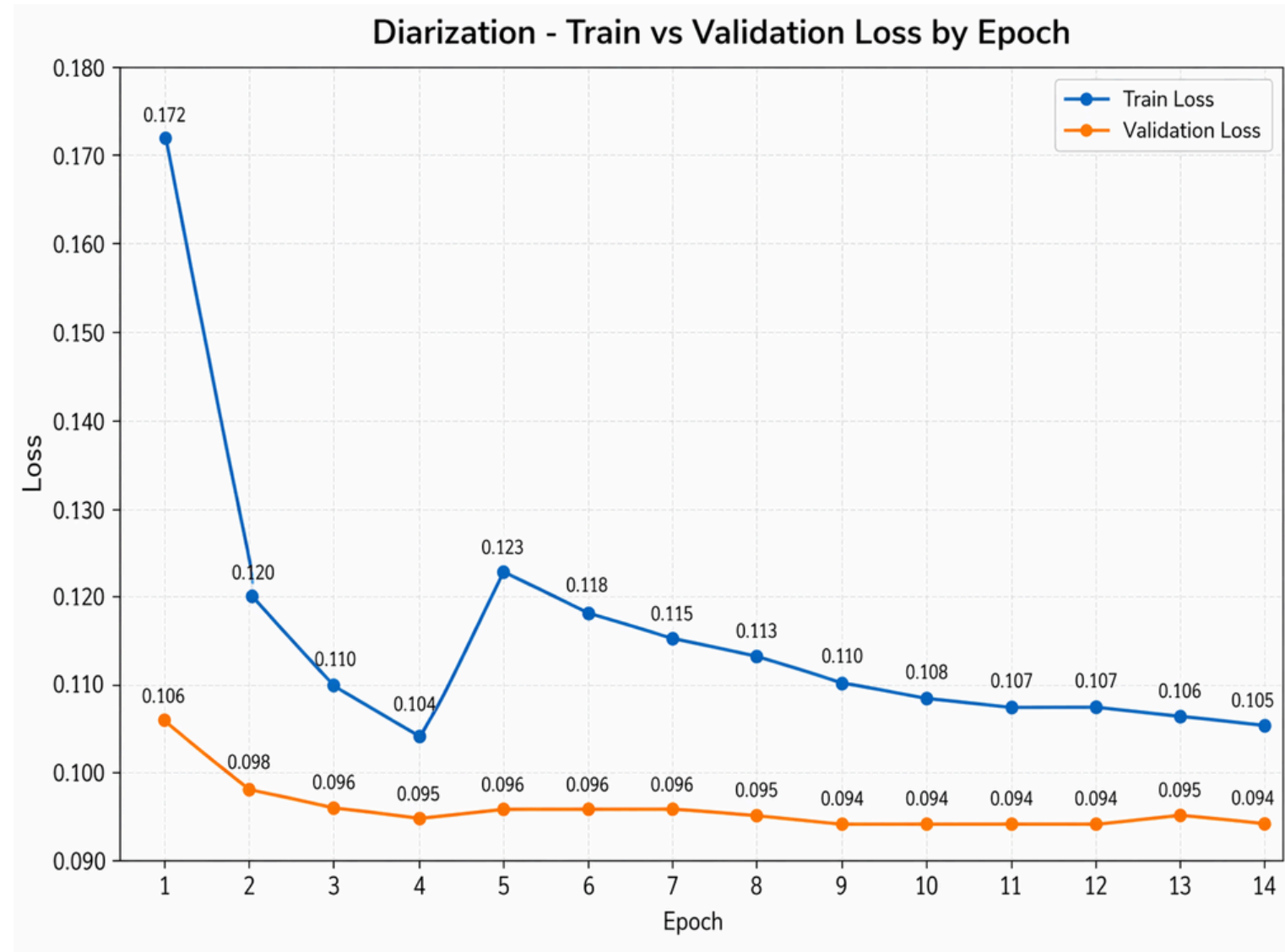
Parameter	Value
Model & Architecture	
Base model	vinai/PhoWhisper-large
Architecture	Whisper Encoder-Decoder
Task	Transcribe (ASR)
Language	Vietnamese (vi)
Sampling rate	16,000 Hz
Fine-tuning Strategy: Partial Unfreezing	
Freeze strategy	Freeze all → unfreeze only q_proj & v_proj
Trainable parameters	q_proj + v_proj across all transformer blocks
Stage 1: Clean Speech Adaptation	
Stage 1: Epochs	2
Stage 1: LR	1e-4
Stage 1: Data mix	~91% Self-labeled + dubbed (100%) / ~9% Synthetic data (10% of base hours)
Stage 1: Overlap filter	≤ 18% overlap ratio per segment
Stage 1: Augmentation	False
Stage 2: Robustness Fine-tuning	
Stage 2: Epochs	1
Stage 2: LR	2e-5 (= 1e-4 × scale 0.2)
Stage 2: Data mix	~95% robust data (Self-labeled + dubbed + robustness 100%) / ~5% Synthetic data (5% of base hours)
Stage 2: Overlap filter	≤ 30% overlap ratio per segment
Stage 2: Augmentation	False
Loss & Metrics	
Loss function	Cross-entropy on token logits
Metric for best checkpoint	Word Error Rate: WER
WER computation	Custom Levenshtein edit-distance (S+D+I)

Parameter	Value
Optimization	
Batch size	2 (per device)
Gradient accumulation	8 steps → effective batch = 16
Optimizer	AdamW
Warmup ratio	0.05
Gradient clipping	1.0
Precision	FP16 or BF16 (mixed precision, auto-detected)
Data Processing	
Min segment duration	2.5s
Max segment duration (preferred)	18.0s
Max segment duration (hard cap)	30.0s
Merge gap	0.5s
Pad left / right	0.20s / 0.30s
Deduplication	2-pass (local per audio + global fingerprint)
Synthetic Data reuse cap	max 2 reuses per base utterance
Validation	
Eval strategy	Per epoch
Save strategy	Per epoch
Save total limit	2 checkpoints
Load best model	True
Max generation length	256 tokens
Data Buckets (CLASS_TO_BUCKET)	
vif	gold_real
conan, dustin, dustin_on_go, chuyen_ho	silver_real
coi_mo, coi_moi	hard_real
Dubbed sources	silver_dubbed
Synthetic Data overlap-augmented	vivos
Robustness sources	robustness

Parameter	Value
Inference	
Decode language	vi
Generation max length	256 tokens
Forced decoder IDs	prefix tokens: language + task
Cache	Disabled
Training History	
Logged to	trainer_state.json (per stage checkpoint dir)

Part 6. Results

6.1 DiariZen — Training Progress



6.1 DiariZen — Threshold Sweep Results

Table 43: DiariZen Fine-tuned DER (%) by threshold on test_labeled set (★ = selected threshold).

Dataset	Metric	thr=0.05	thr=0.15	thr=0.25 ★	thr=0.35	thr=0.45
chuyen_ho	DER (%)	54.996	38.145	32.743	31.793	32.142
	Miss (%)	0.202	1.397	3.890	7.601	11.360
	FA (%)	52.442	34.104	26.220	22.277	19.576
	Conf (%)	2.353	2.644	2.633	1.916	1.206
	Ov.Prec.	0.013	0.027	0.056	0.001	0.000
	Ov.Recall	0.677	0.444	0.225	0.001	0.000
	Ov.F1	0.026	0.051	0.090	0.001	0.000
	Pred.Seg	866	746	713	665	633
	GT Seg	632	632	632	632	632
coi_moi	DER (%)	96.395	62.516	51.035	45.435	43.496
	Miss (%)	0.577	2.901	6.232	10.092	14.871
	FA (%)	91.047	55.242	40.356	31.452	24.623
	Conf (%)	4.771	4.373	4.447	3.892	4.001
	Ov.Prec.	0.028	0.044	0.053	0.062	0.071
	Ov.Recall	0.928	0.710	0.495	0.369	0.240
	Ov.F1	0.055	0.082	0.095	0.107	0.110
	Pred.Seg	1418	1126	1028	959	934
	GT Seg	669	669	669	669	669

dustin	DER (%)	48.716	30.264	24.509	22.686	22.725
	Miss (%)	0.541	1.632	3.398	6.144	9.345
	FA (%)	45.112	25.436	17.707	13.505	10.776
	Conf (%)	3.062	3.197	3.403	3.037	2.604
	Ov.Prec.	0.016	0.022	0.040	0.068	0.126
	Ov.Recall	0.611	0.335	0.261	0.205	0.163
	Ov.F1	0.031	0.042	0.070	0.102	0.142
	Pred.Seg	1127	917	878	870	845
	GT Seg	785	785	785	785	785
vif	DER (%)	8.478	6.986	6.617	6.287	6.341
	Miss (%)	0.052	0.224	0.468	0.743	1.136
	FA (%)	7.968	6.402	5.795	5.308	4.954
	Conf (%)	0.458	0.360	0.355	0.236	0.250
	Ov.Prec.	0.000	0.000	0.000	0.000	0.000
	Ov.Recall	0.000	0.000	0.000	0.000	0.000
	Ov.F1	0.000	0.000	0.000	0.000	0.000
	Pred.Seg	422	410	423	444	473
	GT Seg	625	625	625	625	625

6.1 DiariZen — Threshold Sweep Results

Low-threshold sweep: DER components on representative samples

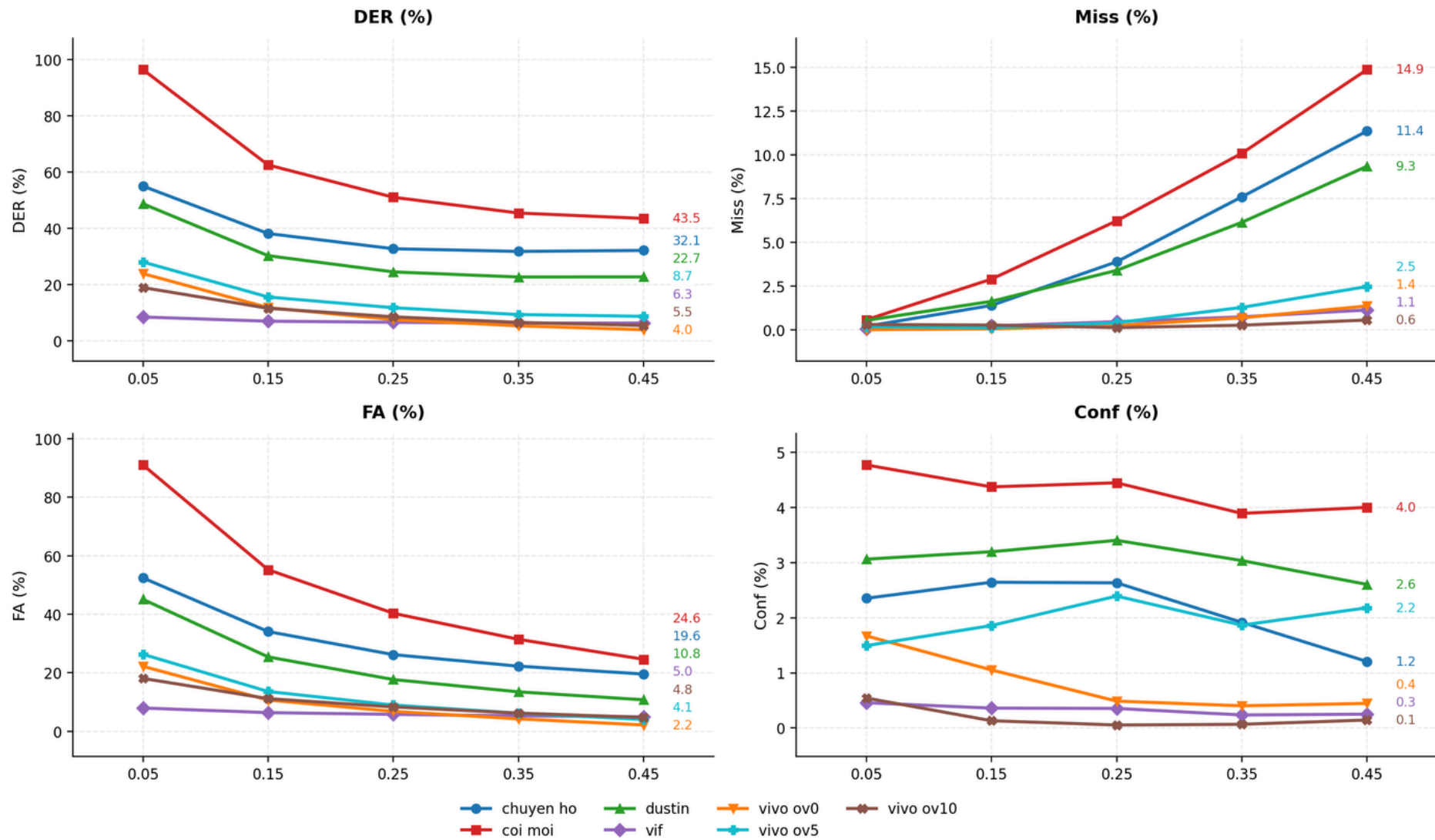


Table 44: DiariZen Fine-tuned DER / Miss / FA / Conf by threshold on Synthetic Data (★ = selected threshold).

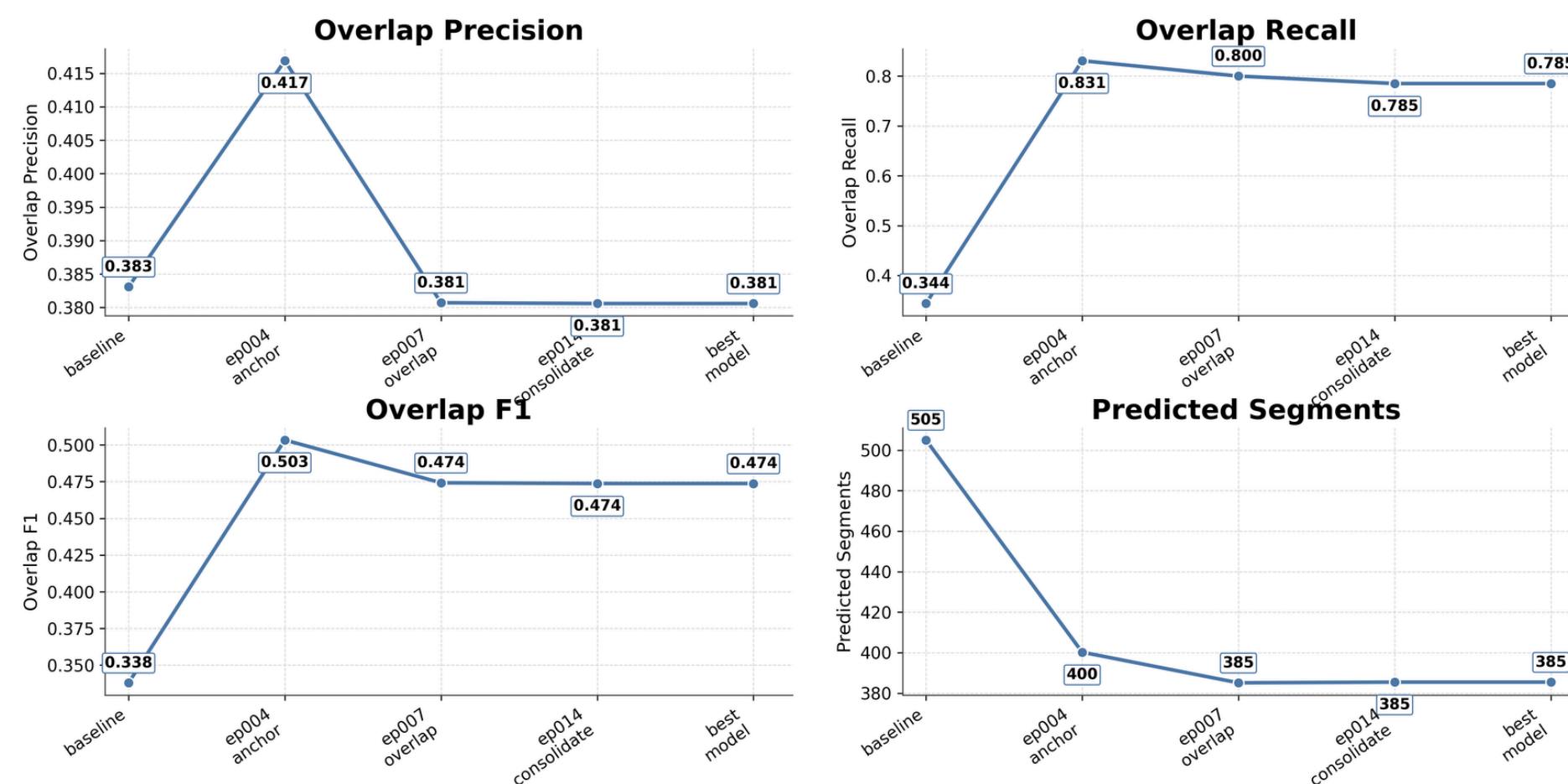
Dataset	Metric	thr=0.05	thr=0.15	thr=0.25 ★	thr=0.35	thr=0.45
Synthetic Data ov0 (GT Seg = 257, no real overlap)						
	DER (%)	23.902	11.818	7.563	5.351	3.980
	Miss (%)	0.011	0.057	0.248	0.689	1.364
	FA (%)	22.220	10.708	6.828	4.260	2.170
	Conf (%)	1.671	1.052	0.487	0.402	0.446
	Ov.Prec.	0.004	0.008	0.009	0.012	0.016
	Ov.Recall	0.949	0.731	0.564	0.410	0.167
	Ov.F1	0.008	0.015	0.018	0.024	0.030
	Pred.Seg	350	321	321	332	323
	GT Seg	257	257	257	257	257
Synthetic Data ov5 (GT Seg = 257, overlap 5%)						
	DER (%)	28.026	15.592	11.779	9.350	8.713
	Miss (%)	0.188	0.123	0.407	1.282	2.476
	FA (%)	26.344	13.610	8.980	6.203	4.057
	Conf (%)	1.493	1.858	2.392	1.864	2.180
	Ov.Prec.	0.220	0.378	0.484	0.584	0.694
	Ov.Recall	0.967	0.993	0.989	0.980	0.966
	Ov.F1	0.358	0.548	0.650	0.732	0.807
	Pred.Seg	448	370	380	375	372
	GT Seg	257	257	257	257	257
Synthetic Data ov10 (GT Seg = 285, overlap 10%)						
	DER (%)	18.943	11.573	8.537	6.554	5.471
	Miss (%)	0.305	0.276	0.128	0.266	0.561
	FA (%)	18.096	11.166	8.355	6.221	4.765
	Conf (%)	0.542	0.131	0.054	0.067	0.144
	Ov.Prec.	0.472	0.598	0.666	0.729	0.778
	Ov.Recall	0.970	0.975	0.997	0.996	0.994
	Ov.F1	0.635	0.741	0.799	0.842	0.873
	Pred.Seg	385	368	365	366	368
	GT Seg	285	285	285	285	285

6.1 DiariZen — Overall Results on VIVOS Mix

Table 45: DiariZen Overall Results on Synthetic Data Set by Checkpoint (thr=0.25).

Checkpoint	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg
Baseline	30.105	15.386	7.927	6.792	0.389	487	276
Phase1	9.641	0.401	7.192	2.048	0.511	400	276
Phase2	13.286	0.650	10.202	2.434	0.483	386	276
Phase3	13.586	1.027	10.224	2.335	0.478	388	276

test_data (vivo) — overall overlap metrics and segment count across checkpoints



6.1 DiariZen — Noise Robustness Evaluation on VIVOS Mix

Table 46: DiariZen DER / Miss / FA / Conf by Condition and Model (thr=0.25).

Condition	Model	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg
0% NOISE								
ov0 / 0% noise	Baseline	30.539	15.943	8.376	6.220	0.000	450	257
ov0 / 0% noise	Best model	7.391	0.248	6.653	0.490	0.018	326	257
ov5 / 0% noise	Baseline	27.654	15.908	7.970	3.777	0.594	489	257
ov5 / 0% noise	Best model	12.378	0.431	9.471	2.475	0.629	388	257
ov10 / 0% noise	Baseline	26.148	14.684	8.451	3.013	0.581	533	285
ov10 / 0% noise	Best model	8.482	0.132	8.295	0.054	0.798	365	285
WITH BACKGROUND NOISE								
ov0 / 23% noise	Baseline	25.882	12.235	7.171	6.477	0.000	370	237
ov0 / 23% noise	Best model	15.368	1.776	11.661	1.931	0.011	328	237
ov5 / 23% noise	Baseline	33.674	17.683	7.738	8.253	0.539	540	294
ov5 / 23% noise	Best model	14.896	1.360	11.463	2.073	0.677	405	294
ov10 / 17% noise	Baseline	36.731	15.865	7.855	13.011	0.617	539	324
ov10 / 17% noise	Best model	23.001	2.215	13.801	6.986	0.734	518	324

6.1 DiariZen — Results by Speaker Count on VIVOS Mix

Table 47: DiariZen DER / Miss / FA / Conf by Speaker Count on Synthetic Data (thr=0.25).

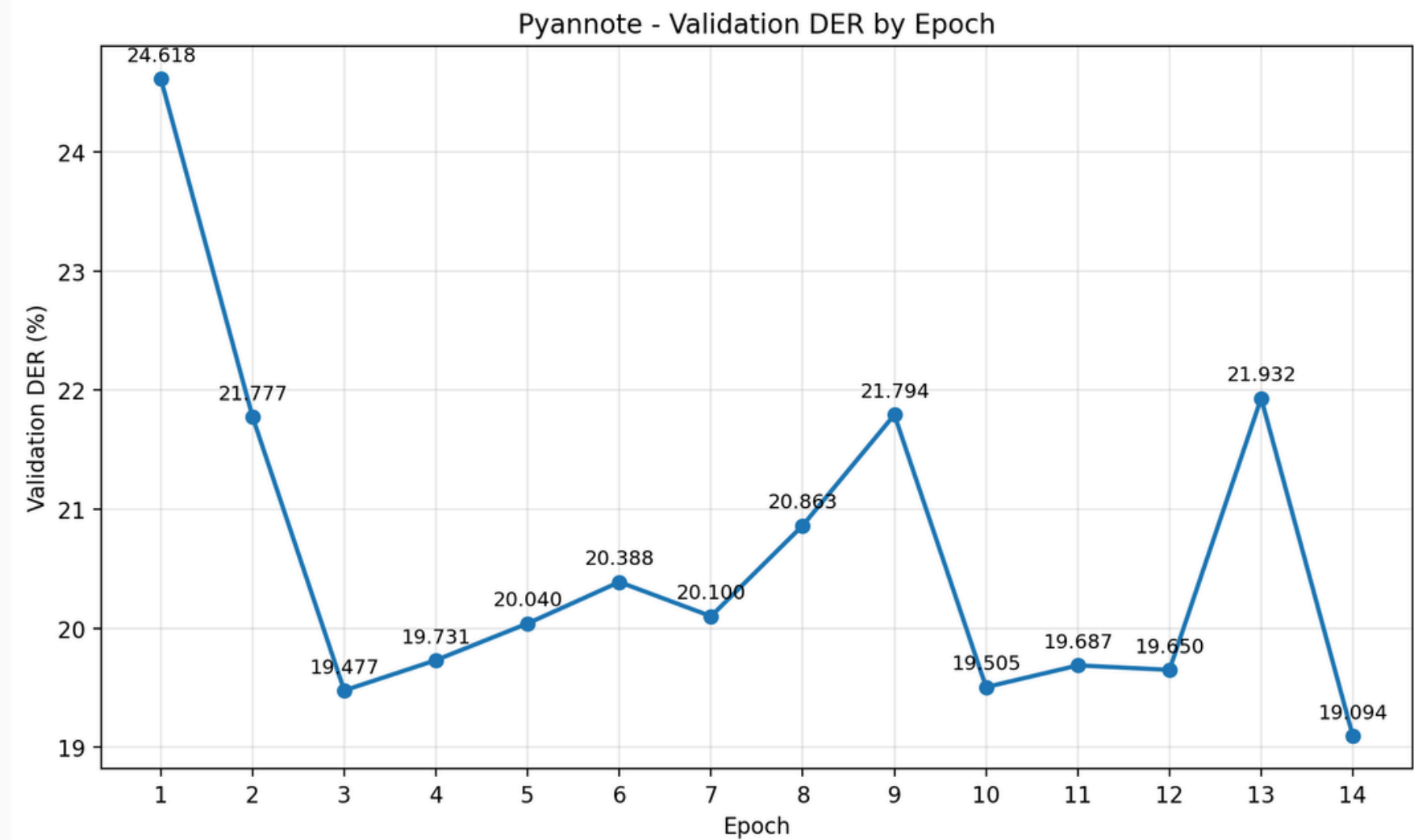
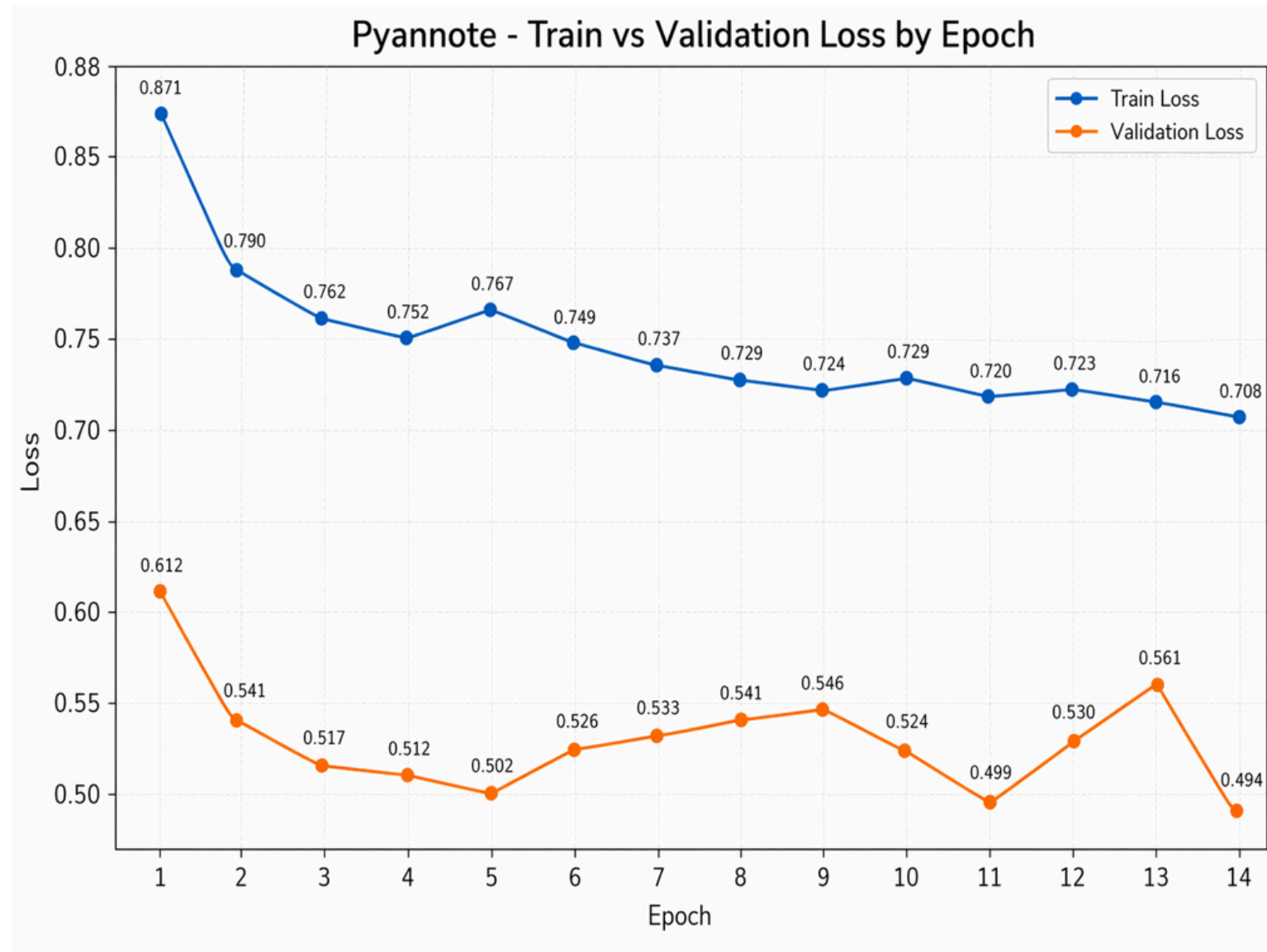
Speakers	Checkpoint	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg
2 spk	Baseline	28.114	15.512	8.266	4.337	0.392	491	266
2 spk	Best model	9.417	0.271	8.140	1.006	0.482	360	266
3 spk	Baseline	31.307	14.050	7.513	9.744	0.309	455	281
3 spk	Best model	19.185	1.995	12.731	4.459	0.373	423	281
4 spk	Baseline	33.674	17.683	7.738	8.253	0.539	540	294
4 spk	Best model	14.896	1.360	11.463	2.073	0.677	405	294

6.1 DiariZen — Results on the Self-Labeled Test Set

Table 48: DiariZen DER / Miss / FA / Conf on test_labeled Set (best model, thr=0.25).

Dataset (test_labeled)	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg	Baseline DER
chuyen_ho	32.742	3.891	26.218	2.633	0.090	713	632	50.801
coi_moi	51.042	6.231	40.364	4.448	0.095	1028	669	72.408
dustin_1	24.516	3.404	17.706	3.405	0.070	879	785	44.078
vif_1	6.617	0.468	5.795	0.355	0.000	423	625	18.248
vif_2	7.521	0.132	7.072	0.317	0.000	318	522	13.404

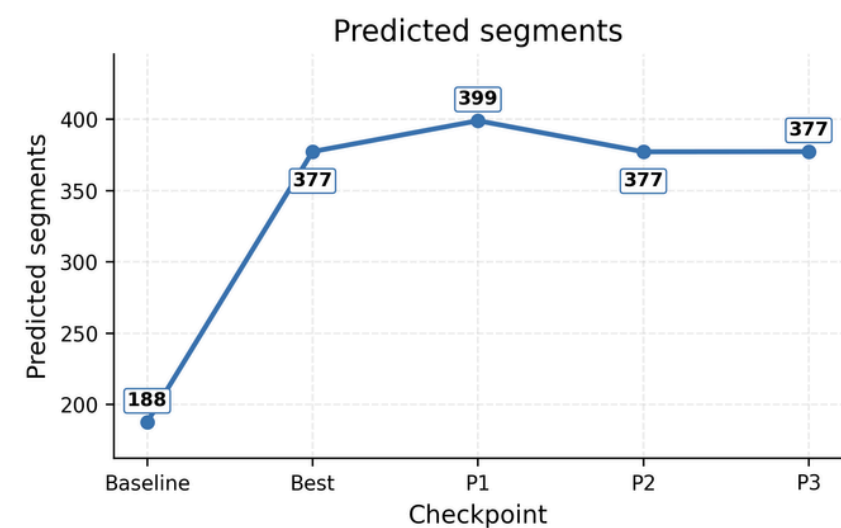
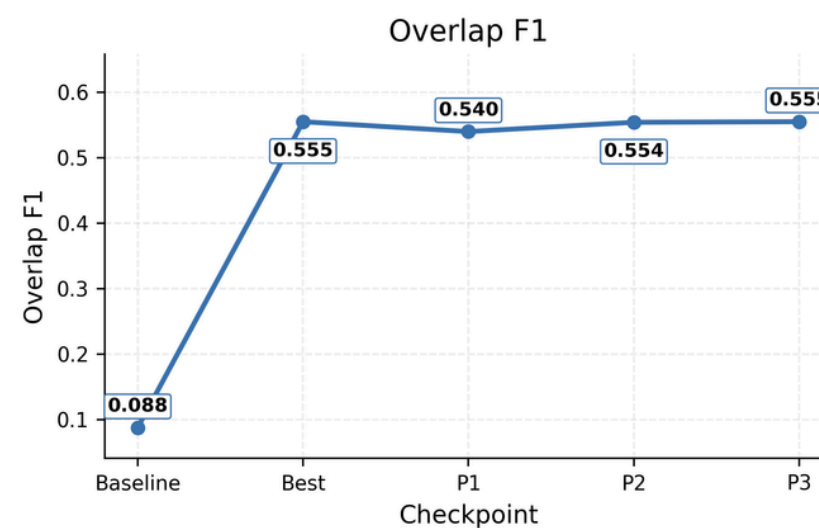
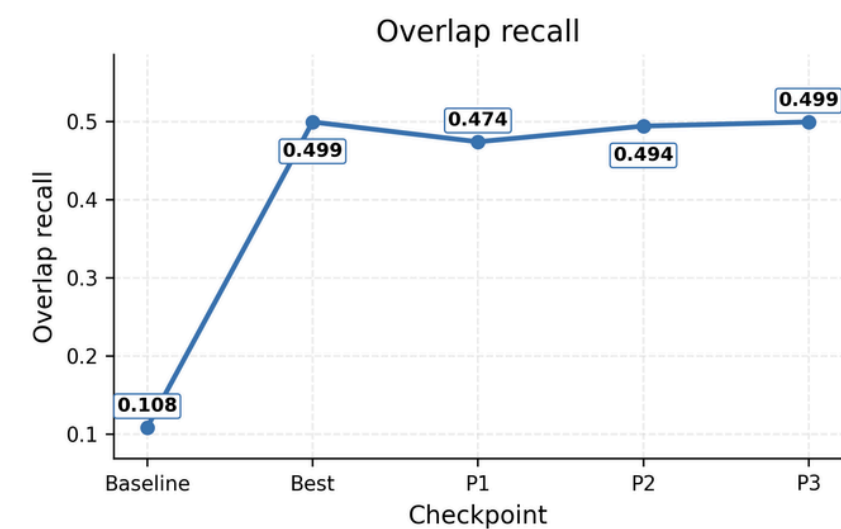
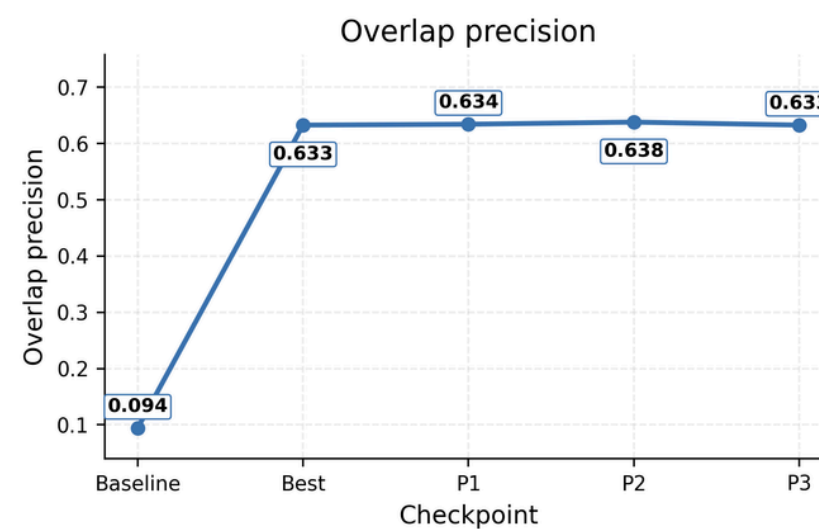
6.2 Pyannote — Training Progress



6.2 Pyannote — Overall Results on VIVOS Mix

Table 49: Pyannote Overall Results on Synthetic Data by Checkpoint.

Checkpoint	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg
Baseline	62.175	20.752	19.574	21.849	0.088	188	271
Phase 1	17.546	5.115	1.372	11.059	0.540	399	271
Phase 2	17.338	4.371	1.742	11.225	0.554	377	271
Phase 3	17.441	4.279	1.761	11.401	0.555	377	271



6.2 Pyannote — Noise Robustness Evaluation on VIVOS Mix

Table 50: Pyannote DER / Miss / FA / Conf by Condition and Model.

Condition	Model	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg
0% NOISE								
ov0 / 0% noise	Baseline	55.974	17.102	17.238	21.634	0.002	173	257
ov0 / 0% noise	Best model	16.121	3.611	1.236	11.274	0.014	352	257
ov5 / 0% noise	Baseline	51.811	16.669	18.727	16.415	0.067	159	257
ov5 / 0% noise	Best model	21.385	2.741	1.391	17.253	0.885	348	257
ov10 / 0% noise	Baseline	57.244	19.722	22.001	15.521	0.177	187	285
ov10 / 0% noise	Best model	3.552	1.697	1.594	0.261	0.940	402	285
WITH BACKGROUND NOISE								
ov0 / 23% noise	Baseline	58.493	17.310	16.001	25.182	0.006	174	237
ov0 / 23% noise	Best model	10.486	2.409	2.898	5.179	0.021	328	237
ov5 / 23% noise	Baseline	66.406	23.583	18.059	24.764	0.156	203	294
ov5 / 23% noise	Best model	18.384	5.576	1.755	11.053	0.778	407	294
ov10 / 17% noise	Baseline	63.952	23.430	18.759	21.762	0.154	210	324
ov10 / 17% noise	Best model	23.171	4.465	1.937	16.769	0.849	460	324

6.2 Pyannote — Results by Speaker Count on VIVOS Mix

Table 51: Pyannote DER / Miss / FA / Conf by Speaker Count on Synthetic Data.

Speakers	Checkpoint	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg
2 spk	Baseline	55.010	17.831	19.322	17.857	0.082	173	266
2 spk	Best model	13.672	2.685	1.411	9.575	0.612	369	266
3 spk	Baseline	67.065	21.226	23.364	22.474	0.087	215	282
3 spk	Best model	17.008	4.490	2.190	10.327	0.520	409	282
4 spk	Baseline	64.450	23.200	16.034	25.216	0.094	175	265
4 spk	Best model	21.643	5.661	1.682	14.300	0.533	355	265

6.2 Pyannote — Results on the Self-Labeled Test Set

Table 52: Pyannote DER / Miss / FA / Conf on test_labeled Set (best model).

Dataset (test_labeled)	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg	Baseline DER
chuyen_ho	41.281	1.895	33.413	5.974	0.066	1042	632	59.164
coi_moi	60.186	6.084	41.661	12.441	0.128	1398	669	92.307
dustin_1	26.412	3.958	16.018	6.436	0.106	1236	785	56.185
vif_1	6.793	2.875	3.618	0.301	0.000	714	625	19.845
vif_2	6.897	0.540	5.972	0.386	0.000	375	522	15.155

6.3 DiariZen vs Pyannote — Noise Robustness Comparison

Table 46: DiariZen DER / Miss / FA / Conf by Condition and Model (thr=0.25).

Condition	Model	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg
0% NOISE								
ov0 / 0% noise	Baseline	30.539	15.943	8.376	6.220	0.000	450	257
ov0 / 0% noise	Best model	7.391	0.248	6.653	0.490	0.018	326	257
ov5 / 0% noise	Baseline	27.654	15.908	7.970	3.777	0.594	489	257
ov5 / 0% noise	Best model	12.378	0.431	9.471	2.475	0.629	388	257
ov10 / 0% noise	Baseline	26.148	14.684	8.451	3.013	0.581	533	285
ov10 / 0% noise	Best model	8.482	0.132	8.295	0.054	0.798	365	285
WITH BACKGROUND NOISE								
ov0 / 23% noise	Baseline	25.882	12.235	7.171	6.477	0.000	370	237
ov0 / 23% noise	Best model	15.368	1.776	11.661	1.931	0.011	328	237
ov5 / 23% noise	Baseline	33.674	17.683	7.738	8.253	0.539	540	294
ov5 / 23% noise	Best model	14.896	1.360	11.463	2.073	0.677	405	294
ov10 / 17% noise	Baseline	36.731	15.865	7.855	13.011	0.617	539	324
ov10 / 17% noise	Best model	23.001	2.215	13.801	6.986	0.734	518	324

Table 50: Pyannote DER / Miss / FA / Conf by Condition and Model.

Condition	Model	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg
0% NOISE								
ov0 / 0% noise	Baseline	55.974	17.102	17.238	21.634	0.002	173	257
ov0 / 0% noise	Best model	16.121	3.611	1.236	11.274	0.014	352	257
ov5 / 0% noise	Baseline	51.811	16.669	18.727	16.415	0.067	159	257
ov5 / 0% noise	Best model	21.385	2.741	1.391	17.253	0.885	348	257
ov10 / 0% noise	Baseline	57.244	19.722	22.001	15.521	0.177	187	285
ov10 / 0% noise	Best model	3.552	1.697	1.594	0.261	0.940	402	285
WITH BACKGROUND NOISE								
ov0 / 23% noise	Baseline	58.493	17.310	16.001	25.182	0.006	174	237
ov0 / 23% noise	Best model	10.486	2.409	2.898	5.179	0.021	328	237
ov5 / 23% noise	Baseline	66.406	23.583	18.059	24.764	0.156	203	294
ov5 / 23% noise	Best model	18.384	5.576	1.755	11.053	0.778	407	294
ov10 / 17% noise	Baseline	63.952	23.430	18.759	21.762	0.154	210	324
ov10 / 17% noise	Best model	23.171	4.465	1.937	16.769	0.849	460	324

6.3 DiariZen vs Pyannote — Results by Speaker Count

Table 47: DiariZen DER / Miss / FA / Conf by Speaker Count on Synthetic Data (thr=0.25).

Speakers	Checkpoint	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg
2 spk	Baseline	28.114	15.512	8.266	4.337	0.392	491	266
2 spk	Best model	9.417	0.271	8.140	1.006	0.482	360	266
3 spk	Baseline	31.307	14.050	7.513	9.744	0.309	455	281
3 spk	Best model	19.185	1.995	12.731	4.459	0.373	423	281
4 spk	Baseline	33.674	17.683	7.738	8.253	0.539	540	294
4 spk	Best model	14.896	1.360	11.463	2.073	0.677	405	294

Table 51: Pyannote DER / Miss / FA / Conf by Speaker Count on Synthetic Data.

Speakers	Checkpoint	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg
2 spk	Baseline	55.010	17.831	19.322	17.857	0.082	173	266
2 spk	Best model	13.672	2.685	1.411	9.575	0.612	369	266
3 spk	Baseline	67.065	21.226	23.364	22.474	0.087	215	282
3 spk	Best model	17.008	4.490	2.190	10.327	0.520	409	282
4 spk	Baseline	64.450	23.200	16.034	25.216	0.094	175	265
4 spk	Best model	21.643	5.661	1.682	14.300	0.533	355	265

6.3 DiariZen vs Pyannote — Self-Labeled Test Set Comparison

Table 48: DiariZen DER / Miss / FA / Conf on test_labeled Set (best model, thr=0.25).

Dataset (test_labeled)	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg	Baseline DER
chuyen_ho	32.742	3.891	26.218	2.633	0.090	713	632	50.801
coi_moi	51.042	6.231	40.364	4.448	0.095	1028	669	72.408
dustin_1	24.516	3.404	17.706	3.405	0.070	879	785	44.078
vif_1	6.617	0.468	5.795	0.355	0.000	423	625	18.248
vif_2	7.521	0.132	7.072	0.317	0.000	318	522	13.404

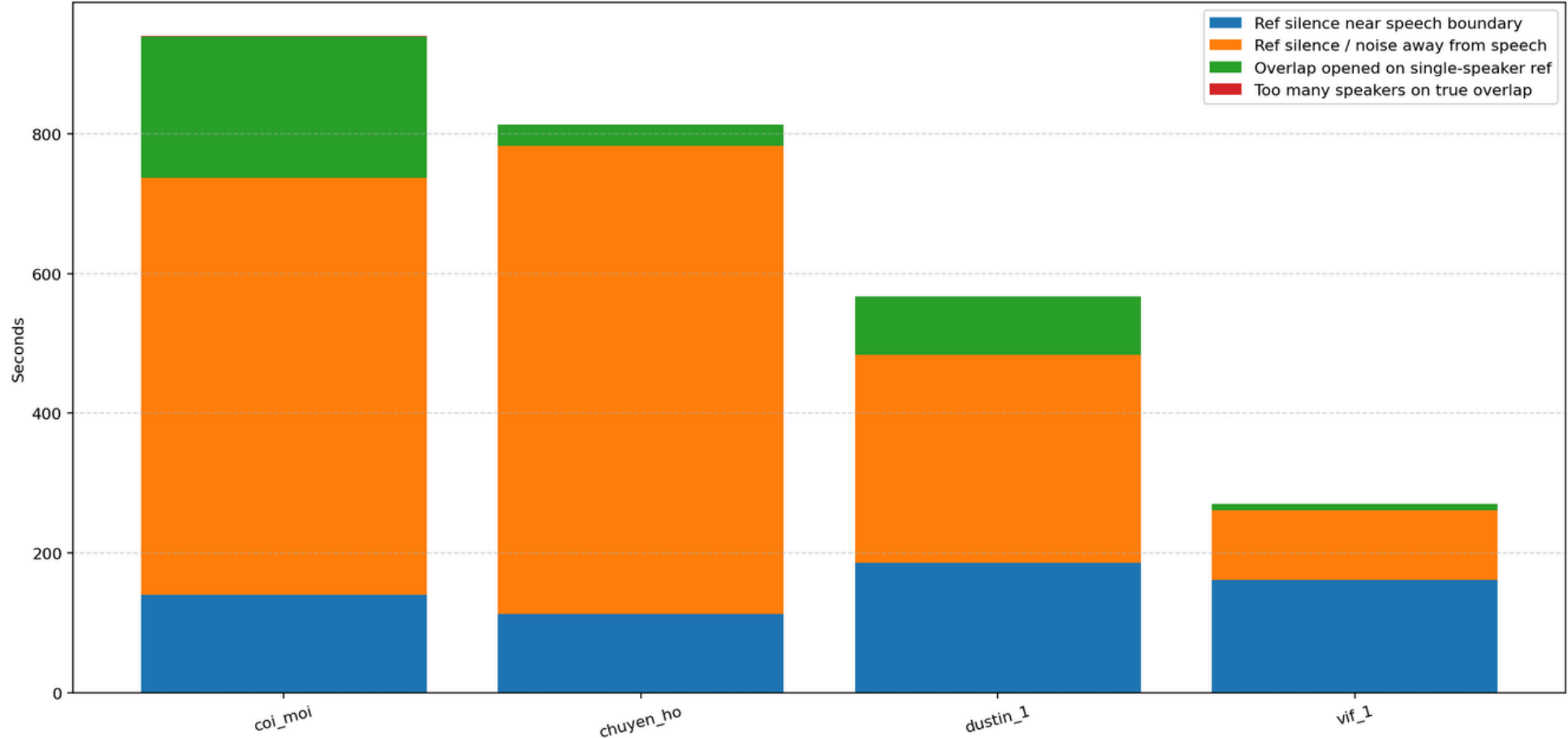
Table 52: Pyannote DER / Miss / FA / Conf on test_labeled Set (best model).

Dataset (test_labeled)	DER(%)	Miss(%)	FA(%)	Conf(%)	Ov.F1	Pred.Seg	GT Seg	Baseline DER
chuyen_ho	41.281	1.895	33.413	5.974	0.066	1042	632	59.164
coi_moi	60.186	6.084	41.661	12.441	0.128	1398	669	92.307
dustin_1	26.412	3.958	16.018	6.436	0.106	1236	785	56.185
vif_1	6.793	2.875	3.618	0.301	0.000	714	625	19.845
vif_2	6.897	0.540	5.972	0.386	0.000	375	522	15.155

6.3 False Alarm Analysis on the Self-Labeled Data

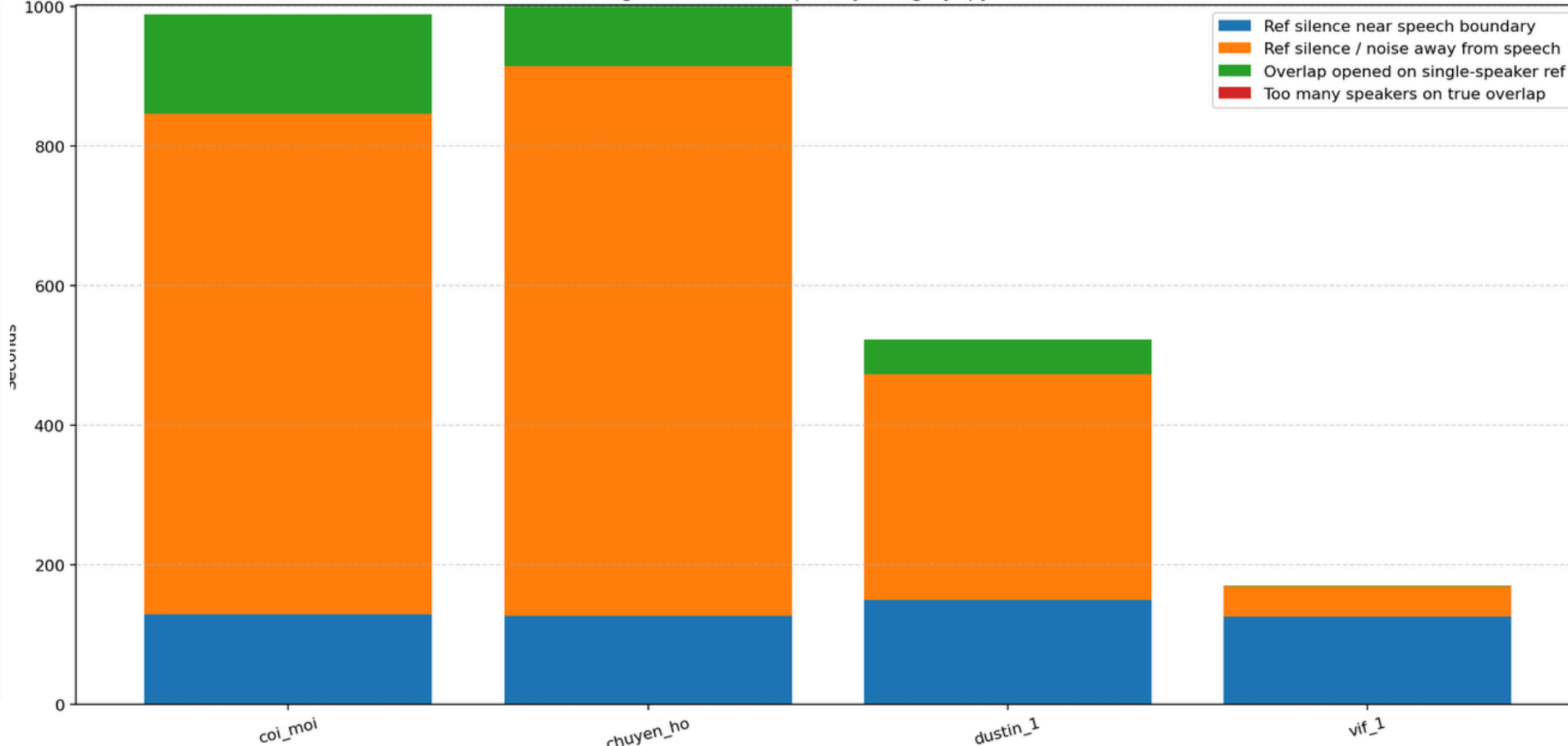
Diarizen

Diagnosed FA-like surplus by category (diarizen)



Pyannote

Diagnosed FA-like surplus by category (pyannote)

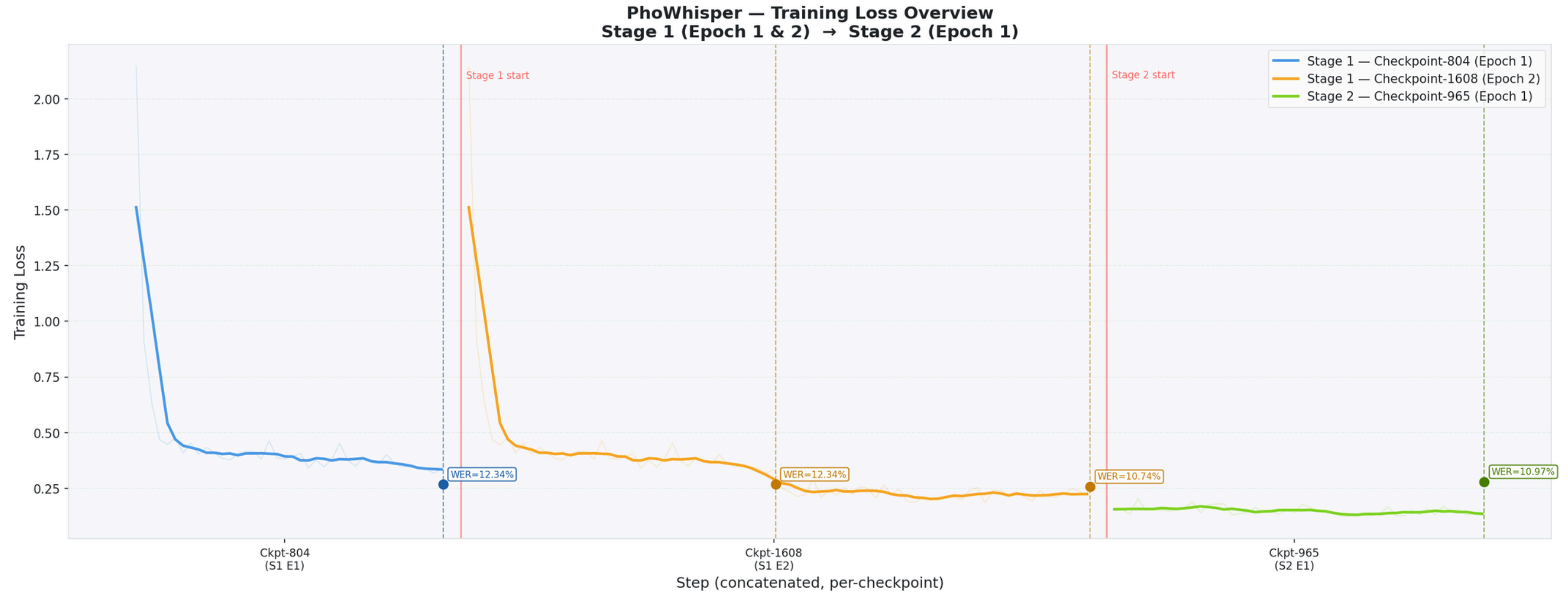


6.3 Final Comparison — DiariZen vs Pyannote

Table 60: Comparison DiariZen vs Pyannote (Best model).

Criterion	Advantage	DiariZen (Best FT)	Pyannote (Best FT)
DER: Synthetic Data (overall)	DiariZen	13.793%	17.441%
Miss: Synthetic Data (overall)	DiariZen	1.138%	4.279%
FA: Synthetic Data (overall)	Pyannote	9.731%	1.761%
Confusion: Synthetic Data (overall)	DiariZen	2.924%	11.401%
Overlap F1: Synthetic Data (overall)	Pyannote	0.478	0.555
Pred. Seg count: Synthetic Data	Pyannote	385	377
DER: 2 speakers	DiariZen	9.438%	13.672%
DER: 3 speakers	Pyannote	17.858%	17.008%
DER: 4 speakers	DiariZen	14.084%	21.643%
Overlap F1: 4 speakers	DiariZen	0.677	0.533
Conf: ov0/0% noise	DiariZen	0.490%	11.274%
FA: ov0/0% noise	Pyannote	6.653%	1.236%
DER: ov0/23% noise	Pyannote	15.368%	10.486%
Loss function	N/A	PIT multilabel BCE	Powerset cross-entropy

6.4 PhoWhisper — Training Loss Overview



6.4 PhoWhisper — Overall WER Progression

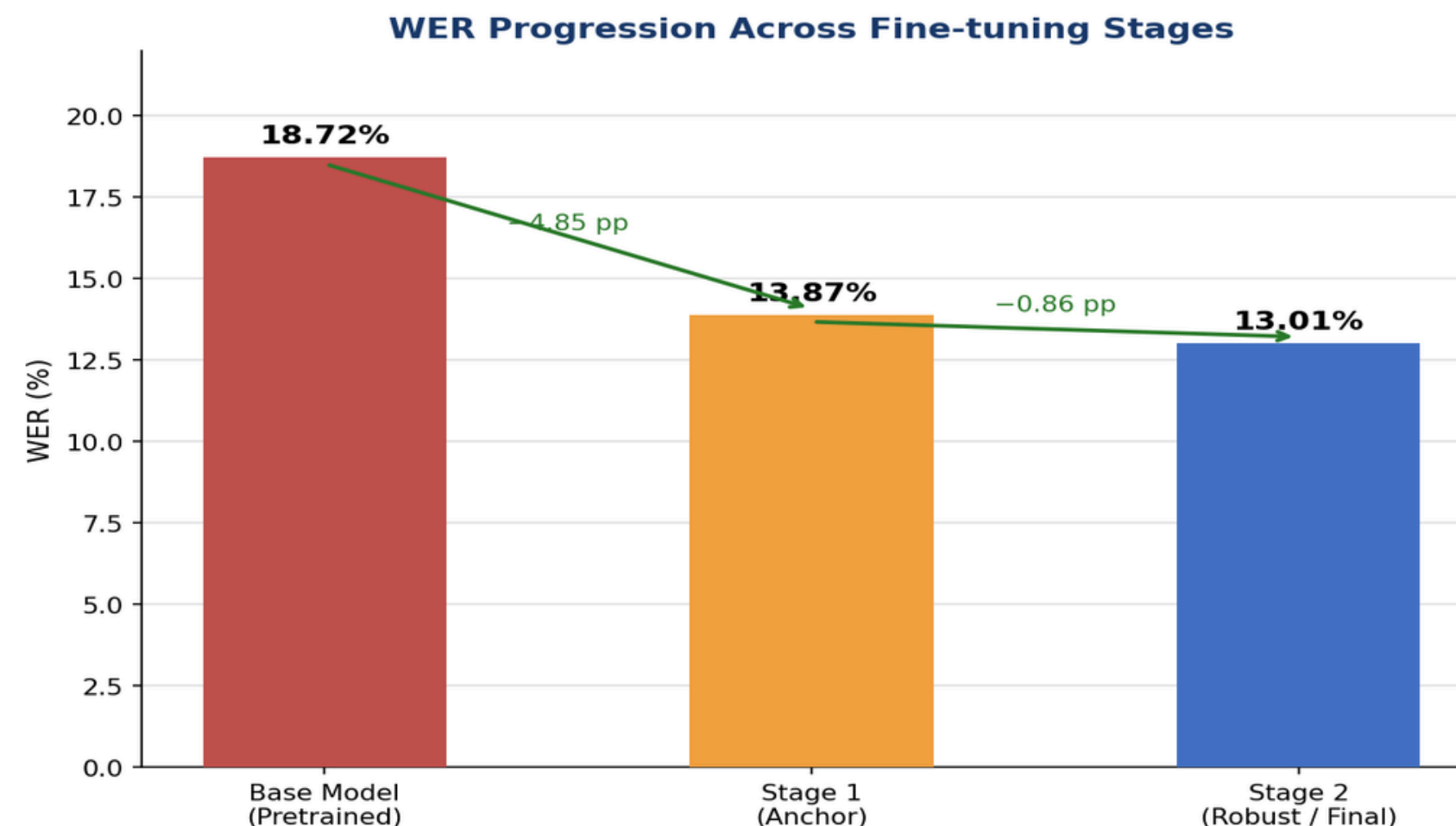


Table 53: Global WER Comparison of Base / Stage 1 / Stage 2 (1,655 test samples).

Metric	Base Model	Stage 1	Stage 2 / Final	Delta
Overall WER (%)	18.72%	13.87%	3.01%	−5.71
Substitutions (S)	3.691	2.743	2.803	−888
Deletions (D)	1.936	2.185	2.110	+174
Insertions (I)	3.589	1.897	1.490	−2.099
S (% of total words)	7.50%	5.57%	5.69%	−1.81
D (% of total words)	3.93%	4.44%	4.29%	+0.36
I (% of total words)	7.29%	3.85%	3.03%	−4.26
Runtime / 1.655 samples	437.68 s	441.32 s	428.12 s	−9.56
Avg (s/sample)	0.2645	0.2667	0.2587	−0.0058

6.4 PhoWhisper — Results by Domain and Quality Bucket

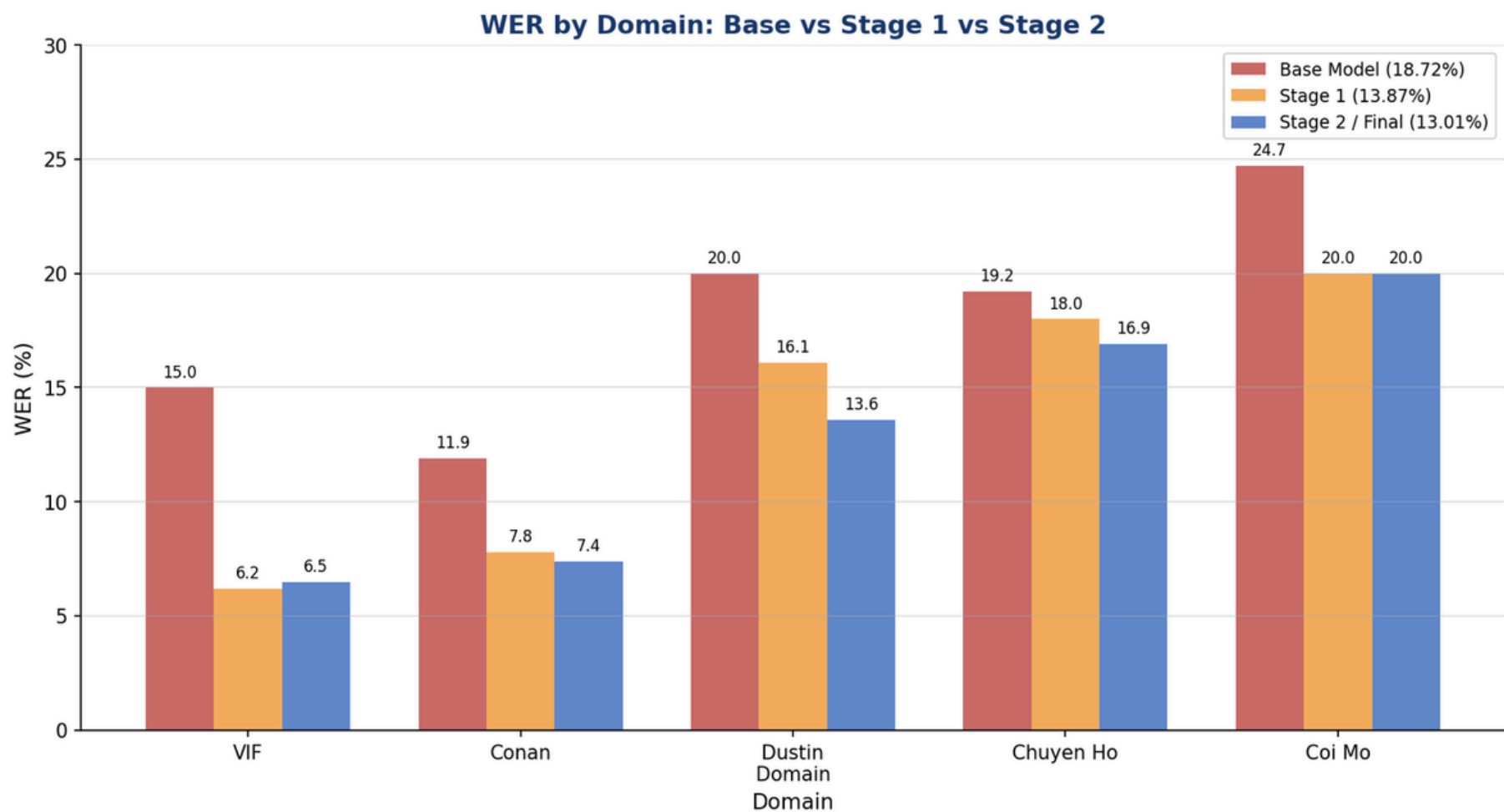


Table 54: WER Comparison of base model and checkpoints.

Domain	Base WER	Stage 1 WER (S/D/I)	Stage 2 WER (S/D/I)	Delta
VIF	14.97%	6.23% (2.38/1.43/2.42)	6.48% (2.5/1.5/2.5)	−8.49
Conan	11.86%	7.75% (3.86/2.79/1.10)	7.38% (3.8/2.7/0.8)	−4.48
Dustin	20.00%	16.12% (5.87/5.09/5.16)	13.62% (5.9/4.9/2.9)	−6.38
Chuyen Ho Chuyen Minh	19.22%	18.03% (6.40/6.12/5.51)	16.93% (6.7/5.9/4.4)	−2.29
Coi Mo	24.73%	19.99% (9.77/6.86/3.36)	19.96% (10.0/6.6/3.4)	−4.77

6.5 Qwen — Automated Evaluation Results Across 5 Datasets

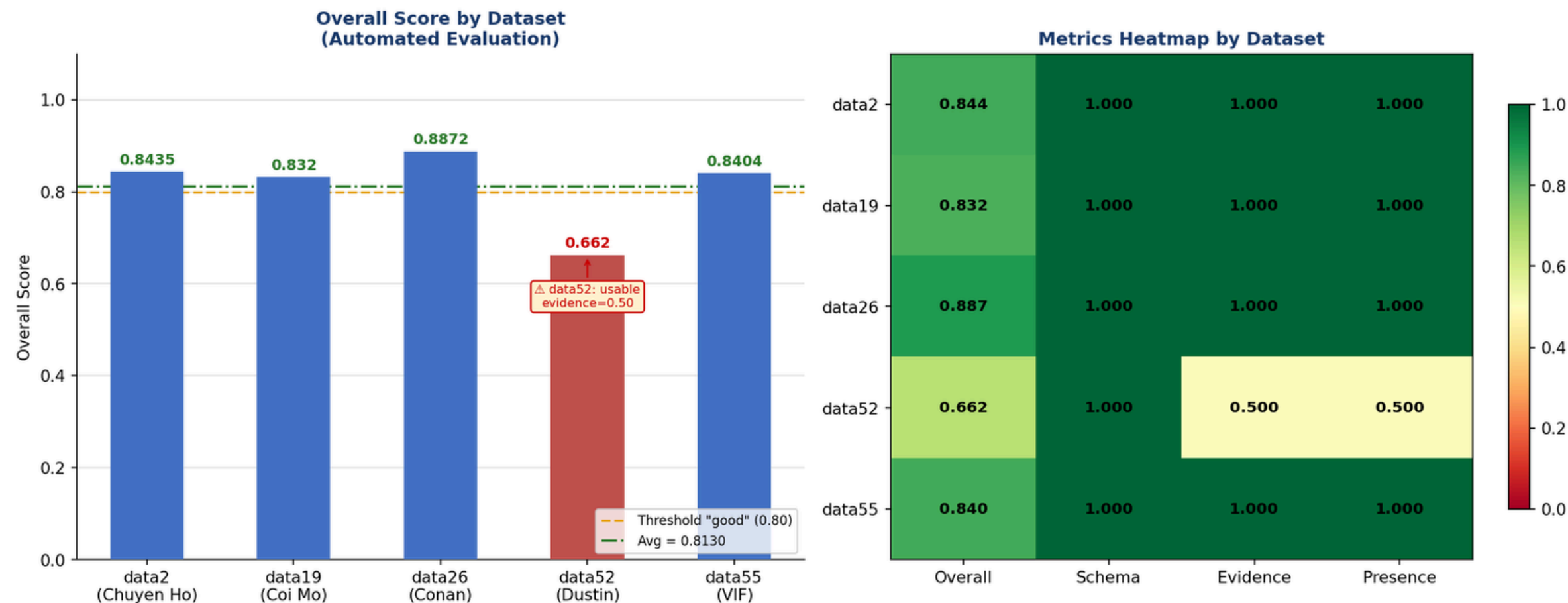


Table 56: Automated Evaluation Results.

Dataset	Overall	Schema	Evidence	Speaker Pres.	Generic	Interp.
Chuyen Ho (Podcast)	0.8435	1.000	1.000	1.000	1.000	good
Coi Mo (Conversation)	0.8320	1.000	1.000	1.000	1.000	good
Conan (Anime)	0.8872	1.000	1.000	1.000	0.615	good
Dustin (Vlog)	0.6620	1.000	0.500	0.500	1.000	usable
VIF (News broadcast)	0.8404	1.000	1.000	1.000	1.000	good
AVG	0.8130	1.000	0.900	0.900	0.923	good

6.5 Qwen — LLM Judge Using Qwen 7B

Table 58: LLM Judge Results Using Qwen 7B.

Criterion	Chuyen Ho	Coi Mo	VIF	DustinGo	Conan	Max Score
Main Summary Score	3	2	2	2	3	4
Speaker Summary Score	2	2	2	1	2	2
Faithfulness Score	2	2	2	3	4	4
Conciseness Score	2	2	2	2	2	2
Total Score	9	8	8	8	11	12

6.5 Qwen — Benchmark

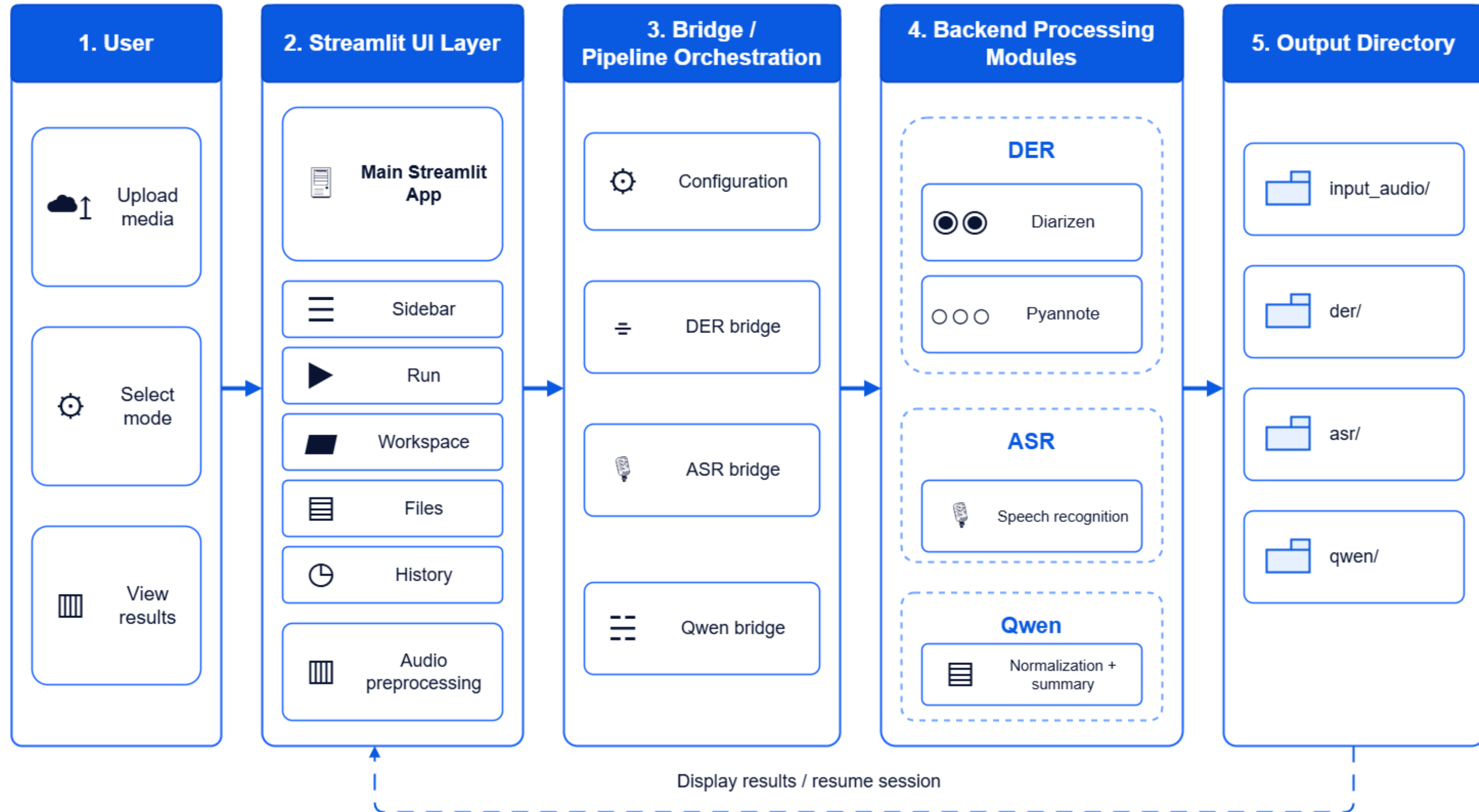
Table 59: VRAM and Tokens/s Benchmark.

Model Variant	Precision	Peak VRAM (MB)	Tokens/s	Notes
Qwen 2.5-1.5B	4-bit	1.196 MB	2,81	Vram less
Qwen 2.5-1.5B	BF16	3.162 MB	12,39	Optimal performance
Qwen 2.5-7B	4-bit	5.740 MB	4,15	Requires 5× more VRAM than 1.5B 4-bit

Part 7: Demo

Part 7 Demo - System design

Architecture Diagram: UI Layer and Backend Processing Modules



Part 7: Demo - Overview UI

DiariZen Pipeline Pro

Refactored Streamlit UI. Backend and pipeline logic are preserved.

Run

Workspace

Files

History

INPUT

Audio input

Upload audio file

Or use a local file path

Upload

200MB per file • WAV, MP3, FLAC

Speaker enrollment

Enrollment source

Do not use

Use an existing local directory

Upload sample audio files (multi-speaker)

PROCESSING OPTIONS

Run mode

Processing scope

Full pipeline (DER → ASR → Qwen)

DER + ASR only (skip Qwen)

Qwen only (use existing ASR results)

Model configuration

Diarization engine

DER checkpoint (.pth)

Maximum speakers

DiariZen (Golden v3)

D:\envs\website\core\der\

4

-

+

Runtime configuration

Audio

-

Enrollment

-

Mode

Qwen only (use existing ASR results)

Qwen

Yes

Execution summary

Selected session

run_20260419_185422

Updated

2026-04-20 10:46:36

Quick status

Complete

DER: Yes

ASR: Yes

Qwen: Yes

Run pipeline

Part 8: Conclusion

Table 62: Summary of achievements and limitations by system component.

Component	Achievements	Limitations
Diarization	Handles noisy multi-speaker audio. Reduces missed speech and keeps speaker confusion relatively low.	when audio has too much noise False alarm is still high. Over-segmentation remains.
ASR	Reduces hallucination after fine-tuning. Produces usable Vietnamese transcripts.	Overlapped speech is not fully handled. The system still struggles with English words in Vietnamese conversations.
Qwen Summarization	Generates local meeting summaries. Works with diarized and transcribed content.	Summaries are still general, and some details may be missed. Because a relatively small local Qwen model is used, its ability to identify the role, responsibility, or contribution of each participant remains limited.
Overall Pipeline	Combines diarization, ASR, and summarization into one framework for Vietnamese meeting analysis.	Needs further improvement in overlap handling, false alarm reduction, ASR robustness, summary detail, and participant-role understanding.